



GENERALIZED ESTIMATION SYSTEM

Version 2.03

Methodology Guide – Item nonresponse and imputation

October 2019

G-Est

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1 Purpose

This document provides the methodology for the variance under imputation in G-EST version 2.03, where G-EST is the new generalized system for estimation developed at Statistics Canada. G-EST replaces three previous StatCan's products, namely GES (Generalized Estimation System), StatMx and SEVANI (System for Estimation of Variance due to Nonresponse and Imputation). The methodology in this current guide is an extension of the methodology guide for SEVANI version 2.3.

The target of this current document is mainly methodologists and any person wanting to know the theory of variance under nonresponse and imputation underlying G-EST. Some background in mathematics and statistics may be required to fully understand this document.

Two other documents will be companions to the current methodology guide:

- The User Guide for G-EST variance module;
- The Methodology Guide of point estimation and variance.

2 Changes

Version 2.03:

There is no change for version 2.03.

Version 2.02:

The main change is the inclusion of skip patterns.

3 Overview

3.1 Sources of errors and variability

Estimates produced by statistical agencies conducting sample surveys are generally subject to many sources of errors. These errors are usually divided into two types: the sampling error and the nonsampling errors. The sampling error occurs because the desired information is collected only for a sample of the whole population. The nonsampling errors may arise for different reasons: errors in the frame, missing values in the sample or measurement errors are examples of nonsampling errors.

G-EST deals with the sampling error and the error resulting from missing values in the sample. The latter is called the nonresponse error in the remaining of this methodology guide. It is assumed, to simplify, that there is no other nonsampling error. It should be noted that the main cause of missing values is usually nonresponse but they may also be introduced during the editing step of the survey due to doubtful or inconsistent responses. The problem raised by missing values is often treated through imputation, especially in business surveys.

The variance and the coefficient of variation of an estimate are two measures frequently used to evaluate its precision. To properly estimate the variance, each source of error has to be taken into account since each source brings its own contribution to the total variability of an estimate. The component of the total variance that is due to the sampling error is called the sampling variance while the component of the total variance that is due to the nonresponse error is called the nonresponse component. In G-EST, the latter is simply defined as the difference between the total variance and the sampling variance. The nonresponse component also includes the imputation variance, if any. The imputation variance is defined as the component of the total variance that results from the use of a random imputation method.

3.2 Levels of units

We consider the general situation where three levels of units exist for the survey data.

The first level is the sampling level, where the sampling design chose the units. This is the most aggregated level. We call units of this level the sampling units.

Then imputation was applied, if any, at a sublevel, called the imputation level. Units of this level are called imputation units. One sampling unit is composed of one or more imputation units.

There is finally a third level, called the estimation level, which is a disaggregation of the imputation level. Such units are called the estimation units. Domain information is at this level. One imputation unit is composed of one or more estimation units.

The passage from the imputation level to the estimation level is done through a process called allocation¹. Numeric values of an imputation unit are distributed among its estimation units using allocation factors. For example, an imputation unit can be made of three estimation units for which the allocation factors are 40%, 35% and 25% respectively. If the value of some Y variable is 100000 for the imputation unit, then the values of this Y variable for the three estimation units will be 40000, 35000 and 25000. It is assumed that the list of estimation units belonging to an imputation unit is known and that the allocation factors are also known and error-free. Therefore, in G-EST, the allocation process does not add a source of variability (see previous section).

This structure of units is the one assumed in G-EST. Of course, as special cases, they cover the situations where all three levels are the same, where the sampling and imputation levels are the same, and where the imputation and estimation levels are the same.

An important example of a survey where the three levels of units are distinct is the Integrated Business Survey Program (IBSP) of Statistics Canada. In that survey, the sampling unit is the enterprise, the imputation unit is the collection entity and the estimation unit is the establishment.

3.3 Estimation and imputation methods

G-EST makes some assumptions.

The first requirement is about the population parameter to be estimated. It can be a domain total, a domain mean or a domain ratio, but in the case of the domain ratio, only the numerator variable can have been subject to nonresponse. In fact, the population parameter can be a generalized total or a generalized ratio, where the first generalizes the domain total while the second is a generalization of both the domain mean and the domain ratio. See chapter 8.

The second assumption is about the form of the estimator of the population parameter. It uses design weights (Horvitz-Thompson estimator) or calibration weights (calibrated estimator).

¹ Not to be confused with sampling allocation.

The third assumption is that imputation was used to compensate for the nonresponse. There must be no missing values remaining in the variables of interest involved in a population parameter, since imputation will have filled all the gaps².

G-EST can cope with three different families of imputation methods: imputation by replacing the missing value by an auxiliary value (AV), sometimes called cold-deck or substitution imputation, linear regression (LR) imputation, which can be deterministic or random (i.e. adding a random residual), and donor imputation, which can be random donor imputation or nearest-neighbour imputation. In fact more than one method can have been used for the same variable of interest and each of these different imputation steps can have used its own set of imputation classes. Imputation methods supported by G-EST are described in greater depth in chapter 6.

3.4 Approach to variance estimation and inference

G-EST is based on an imputation model approach to variance estimation and inference. In this approach, three sources of variability are considered: the imputation model, the sampling design and the nonresponse mechanism. An imputation model is a set of assumptions about the distribution of the variable requiring imputation. Parametric and semi-parametric models can be specified in G-Est. Although this approach depends on the validity of an imputation model, there is no need to specify an explicit nonresponse model, i.e. a model for the unknown nonresponse mechanism. However, it remains necessary to assume that the imputation model that holds for any given population unit continues to hold if one conditions on units that are in the sample or in the set of respondents. It is also assumed that the imputed estimator (that is, the estimator where the true values of the nonrespondents have been replaced by imputed values) of the domain total or mean is approximately unbiased under the imputation model. Note that a fourth source of variability needs to be taken into account when a random residual is added (which for this version of G-EST is only possible in linear regression imputation): the variability due to the random imputation mechanism. Greater detail about the imputation model approach to variance estimation and inference is found in chapter 7.

4 Population and sample

Consider a population that can be described by a hierarchical structure of up to three levels of units. The most disaggregated level is called the estimation level, because this is the level of the domain variables used in population parameters to estimate (which will be described in chapter 8). The most aggregated level is called the sampling level, because the sampling is done

² The exception being units being out of scope or subject to a skip.

at this level (and all units in sublevels are selected, so this is cluster sampling). In-between, there is the imputation level, because imputation, if any, occurs at this level.

Not all surveys need to have three distinct levels. Some surveys may have merged levels. In general, a sampling unit is composed of one or more imputation units, and an imputation unit is composed of one or more estimation units.

An important example where the three levels are distinct is the Integrated Business Survey Program (IBSP) of Statistics Canada, where the sampling unit is the enterprise, the imputation unit is the collection entity and the estimation unit is the establishment.

To represent the population in terms of each level of units, we will use the notations U_S , U_J and U_E for the population of sampling units, the population of imputation units and the population of estimation units, respectively. Also, we will write $U_{J,i}$ to represent the set of imputation units belonging to the sampling unit $i \in U_S$, and we will write $U_{E,k}$ to represent the set of estimation units belonging to the imputation unit $k \in U_J$. This entails that

$$U_J = \bigcup_{i \in U_S} U_{J,i} ,$$

$$U_E = \bigcup_{k \in U_J} U_{E,k} .$$

The subscripts S , J and E will also be used when referring to a numeric variable of interest of the population. Let y be such a variable. Then $y_{S,i}$ is the value of the variable y for the sampling unit $i \in U_S$, $y_{J,k}$ is the value of y for the imputation unit $k \in U_J$ and $y_{E,j}$ is the value of y for the estimation unit $j \in U_E$. Because of the hierarchical nature of the units, we have the following additivity rules:

$$y_{S,i} = \sum_{k \in U_{J,i}} y_{J,k} , \quad i \in U_S ,$$

$$y_{J,k} = \sum_{j \in U_{E,k}} y_{E,j} , \quad k \in U_J .$$

When the imputation level is not equal to the estimation level, the passage from the imputation level to the estimation level is done through a process called allocation³. Numeric values of an imputation unit are distributed among its estimation units using allocation factors. For example, an imputation unit can be made of three estimation units for which the allocation factors are 40%, 35% and 25% respectively. If the value of some Y variable is 100000 for the imputation unit, then the values of this Y variable for the three estimation units will be 40000, 35000 and 25000. It is assumed that the list of estimation units belonging to an imputation unit is known

³ Not to be confused with sampling allocation.

and that the allocation factors are also known and error-free. Therefore, in G-EST, the allocation process does not add a source of variability (see section 3.1).

For some variable of interest y , we will denote f_{yj} the allocation factor for the estimation unit j . This implies

$$y_{\varepsilon,j} = f_{yj} y_{j,k}, \quad j \in U_{\varepsilon,k}, \quad k \in U_j. \quad (4.1)$$

The notation above covers the general case where each variable of interest can have its own set of allocation factors. However, in many surveys where an allocation process took place, it is likely that the allocation factors will be the same for all variables of interest. You may notice that the allocation factors of all estimation units belonging to the same imputation units sum up to 1. In addition, usually each allocation factor is also a number between 0 and 1, but this is not a requirement of the theory for G-EST and so nothing prevents from having some units with a negative allocation factor.

Consider now a sample drawn from the population. Not surprisingly, the notations s_S , s_J and s_ε will be used to represent the sample in terms of sampling units, imputation units and estimation units, respectively. The sampling can be one-phase or two-phase. We define the probability of selection π_i , for $i \in U_S$, as the probability that the sampling unit i belongs to the sample, in the case of one-phase sampling, or as the product of the probabilities to be in the sample in each phase, in the case of two-phase sampling. And we define $w_{S,i}^\circ = 1/\pi_i$, for $i \in s_S$, which can be seen as the design weight. All imputation units and all estimation units belonging to the same sampling unit have the same weight, which we express as

$$w_{S,i}^\circ = w_{J,k}^\circ = w_{\varepsilon,j}^\circ, \quad i \in s_S, \quad k \in U_{J,i}, \quad j \in U_{\varepsilon,k}.$$

5 Out-of-scope units and skip patterns

Sometimes units being part of the sample are considered out of the scope of the survey. These units may still be kept in the sample and be used in calibration, resulting as being relevant in the sampling variance. However they have zeros as values and they are not part of any imputation strategy (chapter 6) or any imputation model (chapter 7). For these units, most of the processing described in the remaining of this document is ignored. In the formulas for inference (chapter 9), all these units have the following characteristics: they are considered respondents for the variable of interest y (hence $r_{yk} = 1$), they have 0 as a value for the variable of interest ($y_{I,k} = 0$), their conditional expectation and variance of $y_{I,k}$ under the imputation model are both zero ($\hat{\mu}_k = 0$, $\hat{\sigma}_k^2 = 0$), they are never involved as contributors to impute nonrespondents (so $\varphi_{kl} = 0$ for all nonrespondents l , and thus $W_{xk} = 0$). Note that the out-of-scope status

should be defined at the imputation unit level or at a higher level (like the sampling unit level). In other words, an imputation unit cannot consist of some in-scope estimation units and some out-of-scope estimation units.

Another type of exclusion happens with skip patterns. A skip pattern is the situation where a variable for a given unit may have a valid missing value. This typically corresponds to a questionnaire where it is said: "If you answered No to Question 3, skip to Question 5. Otherwise please answer Question 4." The variables pertaining to Question 4 are then said to be subject to a skip pattern.

One can see a skip pattern like a partial out-of-scope status. Unlike the out-of-scope expression, it does not apply to all variables of the questionnaire, but only to some.

In G-Est, one out-of-scope expression can be specified, and many different skip pattern expressions can be specified. In fact, for each variable of interest, one can specify the corresponding skip pattern expression, if any. Many variables of interest may share the same skip pattern expression. For example, the Question 4 of the questionnaire can give rise to many variables of interest.

6 Imputation strategy

6.1 List of imputation steps

Each population parameter involves at least one variable of interest, which can have been imputed. The current chapter describes how this imputation can have occurred to be supported by G-EST.

Let y be the variable of interest. In this chapter, we will work mainly with units at the imputation level. Therefore, to alleviate the notation we will drop the index $\mathcal{I}$ to mean the imputation level: we will write s , y_k , etc., instead of $s_{\mathcal{I}}$, $y_{\mathcal{I},k}$, etc. Moreover, we will simply write *unit* instead of imputation unit. Moreover, many quantities defined in this chapter depend on the choice of the variable y , but we won't bother to put a subscript " y " to them even though it is implicitly assumed; hence the notation s_r , in the next paragraph, could be more precisely noted $s_{y,r}$.

The sample s can be divided into two subsets: s_r , which is the list of units where the value of the variable y was responded, and s_m , which is the list of units where the value of the variable y was not responded. Since, as said in section 3.3, all nonresponse must have been treated with imputation, s_m is also the list of units where the value of the variable y was imputed.

We will restrict the use of y_k for the true value of the variable y for the unit k , which is only available if $k \in s_r$. To represent the imputed value of y for a nonrespondent $l \in s_m$, we will use the notation y_l^* . Finally $y_{\bullet k}$ is the value available of the variable y for the unit $k \in s$, that is, $y_{\bullet k}$ is equal to y_k if k is a respondent and is equal to y_k^* if k is a nonrespondent.

It is not unusual that a single imputation method or, more generally, a single imputation step was not enough to impute all missing values for the variable y . Then many imputation steps are applied, where each step is able to impute a subset of the units in s_m . Say that J imputation steps were required. Then we note $s_m^{(j)}$ the subset of s_m that was imputed at step j , $1 \leq j \leq J$. We have $s_m = s_m^{(1)} \cup s_m^{(2)} \cup \dots \cup s_m^{(J)}$. The index j for an imputation step follows the order in which imputation steps were performed. This order, actually, is important. Having run the imputation steps in a different order would have possibly yielded different imputation results.

The differences from an imputation step to the next can be major (a change in the family of the imputation method) or minor (a slight collapsing of imputation classes). They are nevertheless considered distinct imputation steps.

This ordered list of imputation steps, with the complete description of each imputation step, constitutes the imputation strategy for the variable y ⁴.

Note that the term *composite imputation* may be used to denote an imputation strategy with more than one imputation step.

In the next sections, we will describe the imputation methods supported by G-EST. They are grouped into three families. But before doing this, we will present some general concepts and terminology.

6.2 Imputation contributor

We define an *imputation contributor* for an imputation method to be any unit whose y -value can have been used in deriving imputed values for nonrespondents. In the case of linear regression (section 6.8), the imputation contributors are units used to derive the regression beta hat; in the case of donor imputation (section 6.9), the imputation contributors are potential donors.

⁴ A more complete concept would be the ordered list of all imputation steps, for *all* variables of interest, because the imputation of a variable x , say, may have an impact on the imputation later of a variable y . This would be the full imputation strategy, but the theory for the current version of G-EST does not take the full imputation strategy into account. Only the imputation strategy of one variable of interest at a time is supported in this version of G-EST.

In some surveys, only respondents $k \in s_r$ are used as contributors. In other surveys, in addition to respondents, nonrespondents imputed at a previous imputation step may be used as contributors. For an imputation step j , we define

$$s_{R,ALL}^{(j)} = s_r \cup \bigcup_{j'=1}^{j-1} s_m^{(j')}$$

to represent the list of all units having a value for y at the beginning of the imputation step j , albeit only units in s_r have non-imputed values. Then we define $s_R^{(j)}$ the subset of $s_{R,ALL}^{(j)}$ that is the list of imputation contributors used. In imputation steps where previously imputed nonrespondents are not allowed as contributors, the set $s_R^{(j)}$ falls back to s_r (or maybe even a subset of it). In general, we assume there is an imputation weight, noted $\omega_k^{(j),IMP}$, that indicates the imputation contributors:

$$s_R^{(j)} = \{k \in s_{R,ALL}^{(j)} : \omega_k^{(j),IMP} > 0\}.$$

Sometimes, when the number j of the imputation step is assumed, we may simply write s_R to represent the imputation contributors of this step.

Remark 1: The sequence of sets $s_{R,ALL}^{(j)}$, $1 \leq j \leq J$, is increasing: $s_{R,ALL}^{(j)} \subseteq s_{R,ALL}^{(j+1)}$, $1 \leq j \leq J-1$, with $s_{R,ALL}^{(1)} = s_r$.

6.3 Imputation classes

A collection of imputation classes for an imputation step is a partitioning of the sample such that the value imputed to the variable of interest for some nonrespondent can only be built from units coming from the same imputation class as the nonrespondent.

Mathematically

$$y_l^* = \text{function}(s_c, l)$$

where l is a nonrespondent from the imputation class c and s_c is the portion of the sample belonging to the imputation class c .

Examples of collections of imputation classes are:

- All provinces and territories in Canada;
- All NAICS6 codes⁵;

⁵ NAICS: North-Atlantic Industrial Classification System

- Cross-product of the provinces/territories and the NAICS3 codes.

Each imputation step can have its own collection of imputation classes. Generally we will always assume that imputation is done with a collection of imputation classes, even though this collection can be a dummy one (only one imputation class corresponding to the full sample).

A note of caution here: when we write $y_l^* = f(s_c, l)$, it means the sample at its state at the beginning of the imputation step. When the step is not the first one, it means that (possibly) some imputation already occurred and this imputation may have used different collections of imputation classes. If the current imputation step allows using previously imputed values for its imputation contributors, and some of these nonrespondents were imputed in steps using other collections of imputation classes, then in fact y_l^* does not depend only on the responded values of s_c , but maybe also of other respondents as well.

Let c be an imputation class from the collection of imputation classes. We will sometimes make use of the following notations:

$s_{m,c}$ or $s_{m,c}^{(j)}$	list of nonrespondents belonging to the class c ,
$s_{r,c}$ or $s_{r,c}^{(j)}$	list of respondents belonging to the class c ,
$s_{R,c}$ or $s_{R,c}^{(j)}$	list of imputation contributors belonging to the class c .

6.4 Deterministic and stochastic imputation

Let y be a variable with nonresponse.

When the imputed value to a nonrespondent $l \in s_m$ only depends on the observed values in the sample and of no external stochastic mechanism, we say that the imputation is *deterministic*. In other words, it is repeatable with the same results, given the same sample. The imputed value can be derived as $y_l^* = f(s, l)$, where f is some known function using values from the sample.

On the other hand, when the imputed value depends also on some random mechanism (e.g. throwing dice to select among values to be used as imputed value), the imputation is said *stochastic* or *random*. The results of imputation are not repeatable, unless this external random mechanism is controlled (e.g. with a fixed seed for pseudo-random numbers).

One way to provide stochastic imputation is to start with a deterministic imputation, and then add a random residual. The value issued from the deterministic imputation is called the *initial imputed value* and is noted y_l^{o*} for the nonrespondent $l \in s_m$. The random residual, noted e_l^* , is a residual chosen in some stochastic way among the residuals e_k of the imputation contributors $k \in s_R$, and may be adjusted (with a variance structure, or with other criteria). The residual of

an imputation contributor k is defined as the value⁶ $y_{\bullet k}$ less the value it would have been imputed by the deterministic imputation at the step, $y_k^* = f(s, k)$, if that imputation was applied. Thus the residual of the respondent k is $e_k = y_{\bullet k} - f(s, k)$. In the case of stochastic imputation, one has $y_l^* = y_l^{\circ*} + e_l^* = f(s, l) + e_l^*$.

Stochastic imputation is attractive because it helps recreate variability in the imputed values to be more similar to the variability of the true values of the variable y .

We will write $I_{\text{RESID},l} = 1$ to say that a random residual has been added to the initial imputed value, and $I_{\text{RESID},l} = 0$ if no random residual has been added. When no random residual is added, the initial imputed value is the same as the (final) imputed value, of course. In general, we will write

$$y_l^* = y_l^{\circ*} + e_l^*$$

that allows to encompass in the same expression both deterministic and stochastic imputation, where e_l^* is a random residual if $I_{\text{RESID},l} = 1$ or is equal to zero if $I_{\text{RESID},l} = 0$.

Note we do the following assumptions about the residual e_l^* :

1. $E_{\star}(e_l^*) = 0$;
2. $V_{\star}(e_l^*) = \sigma_l^2$;
3. $\text{cov}_{\star}(e_k^*, e_l^*) = 0$, for $k \neq l$.

The subscript $\star$ indicates that the expectation, variance and covariance are taken with respect to the random imputation mechanism (“throwing the dice”). The quantity σ_l^2 is defined in section 7.2. It is an estimate of the error variance of the imputation model underlying the imputation method. The reasons justifying the above properties is to ensure that imputed values do not destroy the original distribution of the variable y . For instance, e_l^* could be obtained as $e_l^* = \hat{\sigma}_l(e_k - E_{\star}(e_k))/\sqrt{V_{\star}(e_k)}$, where $e_k = (y_k - y_l^{\circ*})/\hat{\sigma}_k$ and k is a randomly selected respondent (within imputation class c or not).

6.5 Linearity of the imputation methods

A feature shared by all imputation methods supported by G-EST is linearity in terms of y -values of the imputation contributors.

We say that an imputation method is linear if the non-stochastic part of the imputed value y_l^* can be expressed as the linear combination

⁶ If the imputation contributor is a previously imputed nonrespondent, the true value y_k must be replaced by its imputed value from a previous step.

$$y_l^{\circ*} = \varphi_{0l} + \sum_{k \in s_r} \varphi_{kl} y_k, \quad l \in s_m^{(\cdot)}, \quad (6.1)$$

where $s_m^{(\cdot)}$ is the subset of s_{ym} that fulfills the conditions to be imputable by the imputation method (i.e. no missing value for the auxiliary variables of the method) and φ_{0l} and φ_{kl} are quantities that are obtained without using y -values. While the indices k and l in φ_{0l} and φ_{kl} seem to suggest only one or two units were used to derive their values, in fact these two quantities may depend on the full sample (respondents and nonrespondents alike), using auxiliary variables, imputation classes, and so on, as long as they do not depend on the values of the variable y .

Such an expression of the imputed value is appealing because it allows to separate in a convenient form the variable of interest from auxiliary variables, the latter which will be seen as fixed when conditioning under the imputation model (see chapters 7 [Imputation model] and section 8.1[Population parameters]).

Now consider the general context of an imputation strategy composed of J imputation steps for the variable of interest y . If none of the J imputation steps allows use of previously y -imputed nonrespondents as imputation contributors, then all the imputation methods supported by G-EST (AV imputation [§ 6.7], LR imputation [§ 6.8], donor imputation[§ 6.9]) support directly the linear condition.

But suppose the case where previously y -imputed nonrespondents were used as imputation contributors in some imputation steps, then things get a bit more complex and the form (6.1) is not straightforward.

So consider that for each imputation step j , the imputed value y_l^* , for $l \in s_m^{(j)}$, can be expressed by the linear combination:

$$y_l^* = \tilde{\varphi}_{0l}^{(j)} + \sum_{k \in s_R^{(j)}} \tilde{\varphi}_{kl}^{(j)} y_{\bullet k} + \tilde{e}_l^*, \quad l \in s_m^{(j)}, \quad (6.2)$$

where $\tilde{\varphi}_{0l}^{(j)}$ and $\tilde{\varphi}_{kl}^{(j)}$ are quantities that are obtained without using y -values and $\tilde{e}_l^*$ is either a random residual chosen among units in $s_R^{(j)}$ when $I_{\text{RESID},l} = 1$ or is defined to be 0 when $I_{\text{RESID},l} = 0$. Note that $\tilde{\varphi}_{0l}^{(j)}$ and $\tilde{\varphi}_{kl}^{(j)}$ are defined only for $l \in s_m^{(j)}$ and $k \in s_R^{(j)}$ (or, equivalently, they are zero elsewhere). Notice that the deterministic part of (6.2) differs from (6.1) in that the sum is done over the imputation contributors, which may include nonrespondents as well as respondents.

Sometimes (6.2) can be rewritten as

$$y_l^* = \tilde{\varphi}_{0l}^{(j)} + \sum_{k \in s_{R,ALL}^{(j)}} \tilde{\varphi}_{kl}^{(j)} y_{\bullet k} + \tilde{e}_l^*, \quad l \in s_m^{(j)}, \quad (6.3)$$

by setting $\tilde{\varphi}_{kl}^{(j)} = 0$ for $k \in s_{R,ALL}^{(j)} - s_R^{(j)}$. However, in the form (6.3), we still have $\tilde{e}_l^*$ being a random residual picked from $s_R^{(j)}$ only (when $I_{RESID,l} = 1$).

The sum in (6.2) is done over $s_R^{(j)}$, but it may happen that for a specific nonrespondent $l \in s_m^{(j)}$, the coefficient $\tilde{\varphi}_{kl}^{(j)}$ is 0 for some contributors $k \in s_R^{(j)}$. In fact, for an imputation method using imputation classes, the value of $\tilde{\varphi}_{kl}^{(j)}$ is 0 whenever k is not in the same imputation class as l .

The form (6.2) is quite common and for the families of imputation methods supported by G-EST (see sections 6.7, 6.8 and 6.9), the exact expressions for $\tilde{\varphi}_{0l}$ and $\tilde{\varphi}_{kl}$ are easily obtained.

However for the sake of variance estimation in section 8.4, we need to rewrite the imputation methods of all the imputation steps so that they depend on the y -values only through respondents. The following proposition achieves this result.

Proposition:

If each imputation step of the imputation strategy of a variable y satisfies (6.2) or, alternatively, (6.3), then they also satisfy

$$y_l^* = \varphi_{0l}^{(j)} + \sum_{k \in s_r} \varphi_{kl}^{(j)} y_k + e_l^*, \quad l \in s_m^{(j)}, 1 \leq j \leq J, \quad (6.4)$$

with the following recursive formulas:

$$\varphi_{0l}^{(1)} = \tilde{\varphi}_{0l}^{(1)}, \quad \varphi_{kl}^{(1)} = \tilde{\varphi}_{kl}^{(1)}, \quad e_l^* = \tilde{e}_l^*, \quad k \in s_r, \quad l \in s_m^{(1)}, \quad (6.5)$$

$$\varphi_{0l}^{(j)} = \tilde{\varphi}_{0l}^{(j)} + \sum_{j'=1}^{j-1} \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \varphi_{0l'}^{(j')}, \quad l \in s_m^{(j)}, 2 < j \leq J, \quad (6.6)$$

$$\varphi_{kl}^{(j)} = \tilde{\varphi}_{kl}^{(j)} + \sum_{j'=1}^{j-1} \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \varphi_{kl'}^{(j')}, \quad l \in s_m^{(j)}, k \in s_r, 2 < j \leq J \quad (6.7)$$

and

$$e_l^* = \tilde{e}_l^* + \sum_{j'=1}^{j-1} \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} e_{l'}^*, \quad l \in s_m^{(j)}, k \in s_r, 2 < j \leq J. \quad (6.8)$$

Proof:

The proof will be done by recurrence.

The case for $j=1$ is obvious, since there is no previously imputed data that can be used. Note that $s_R^{(1)} = s_r$.

Before doing the induction hypothesis, let's just have a look for the cases $j = 2$ and 3.

At imputation step 2:

Let $l \in s_m^{(2)}$. Remark that $s_R^{(2)} = s_r \cup s_m^{(1)}$.

$$\begin{aligned}
 y_l^* &= \tilde{\varphi}_{0l}^{(2)} + \sum_{k \in s_R^{(2)}} \tilde{\varphi}_{kl}^{(2)} y_{\cdot k} + \tilde{e}_l^* \\
 &= \tilde{\varphi}_{0l}^{(2)} + \sum_{k \in s_r} \tilde{\varphi}_{kl}^{(2)} y_k + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(2)} y_{l'}^* + \tilde{e}_l^* \\
 &= \tilde{\varphi}_{0l}^{(2)} + \sum_{k \in s_r} \tilde{\varphi}_{kl}^{(2)} y_k + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(2)} \left[\tilde{\varphi}_{0l'}^{(1)} + \sum_{k \in s_r} \tilde{\varphi}_{kl'}^{(1)} y_k + \tilde{e}_{l'}^* \right] + \tilde{e}_l^* \\
 &= \tilde{\varphi}_{0l}^{(2)} + \sum_{k \in s_r} \tilde{\varphi}_{kl}^{(2)} y_k + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(2)} \left[\varphi_{0l'}^{(1)} + \sum_{k \in s_r} \varphi_{kl'}^{(1)} y_k + e_{l'}^* \right] + \tilde{e}_l^* \\
 &= \left\{ \tilde{\varphi}_{0l}^{(2)} + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(2)} \varphi_{0l'}^{(1)} \right\} + \sum_{k \in s_r} \left[\tilde{\varphi}_{kl}^{(2)} + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(2)} \varphi_{kl'}^{(1)} \right] y_k + \left\{ \tilde{e}_l^* + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(2)} \tilde{e}_{l'}^* \right\} \\
 &= \varphi_{0l}^{(2)} + \sum_{k \in s_r} \varphi_{kl}^{(2)} y_k + e_l^* .
 \end{aligned}$$

Now let $l \in s_m^{(3)}$.

$$\begin{aligned}
 y_l^* &= \tilde{\varphi}_{0l}^{(3)} + \sum_{k \in s_r} \tilde{\varphi}_{kl}^{(3)} y_k + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(3)} y_{l'}^* + \sum_{l' \in s_m^{(2)}} \tilde{\varphi}_{l'l}^{(3)} y_{l'}^* + \tilde{e}_l^* \\
 &= \tilde{\varphi}_{0l}^{(3)} + \sum_{k \in s_r} \tilde{\varphi}_{kl}^{(3)} y_k + \sum_{l' \in s_m^{(1)}} \tilde{\varphi}_{l'l}^{(3)} \left[\varphi_{0l'}^{(1)} + \sum_{k \in s_r} \varphi_{kl'}^{(1)} y_k + e_{l'}^* \right] \\
 &\quad + \sum_{l' \in s_m^{(2)}} \tilde{\varphi}_{l'l}^{(3)} \left[\varphi_{0l}^{(2)} + \sum_{k \in s_r} \varphi_{kl}^{(2)} y_k + e_{l'}^* \right] + \tilde{e}_l^*
 \end{aligned}$$

$$\begin{aligned}
&= \left\{ \tilde{\varphi}_{0l}^{(3)} + \sum_{l' \in S_m^{(1)}} \tilde{\varphi}_{l'l}^{(3)} \varphi_{0l'}^{(1)} + \sum_{l' \in S_m^{(2)}} \tilde{\varphi}_{l'l}^{(3)} \varphi_{0l'}^{(2)} \right\} \\
&\quad + \sum_{k \in S_r} \left[\tilde{\varphi}_{kl}^{(3)} + \sum_{l' \in S_m^{(1)}} \tilde{\varphi}_{l'l}^{(3)} \varphi_{kl'}^{(1)} + \sum_{l' \in S_m^{(2)}} \tilde{\varphi}_{l'l}^{(3)} \varphi_{kl'}^{(2)} \right] y_k \\
&\quad + \left\{ \tilde{e}_l^* + \sum_{l' \in S_m^{(1)}} \tilde{\varphi}_{l'l}^{(3)} e_{l'}^* + \sum_{l' \in S_m^{(2)}} \tilde{\varphi}_{l'l}^{(3)} e_{l'}^* \right\} \\
&= \left\{ \tilde{\varphi}_{0l}^{(3)} + \sum_{j'=1}^{3-1} \sum_{l' \in S_m^{(j')}} \tilde{\varphi}_{l'l}^{(3)} \varphi_{0l'}^{(j')} \right\} + \sum_{k \in S_r} \left[\tilde{\varphi}_{kl}^{(3)} + \sum_{j'=1}^{3-1} \sum_{l' \in S_m^{(j')}} \tilde{\varphi}_{l'l}^{(3)} \varphi_{kl'}^{(j')} \right] y_k \\
&\quad + \left\{ \tilde{e}_l^* + \sum_{j'=1}^{3-1} \sum_{l' \in S_m^{(j')}} \tilde{\varphi}_{l'l}^{(3)} e_{l'}^* \right\} \\
&= \varphi_{0l}^{(3)} + \sum_{k \in S_r} \varphi_{kl}^{(3)} y_k + e_l^* .
\end{aligned}$$

Now let's assume the recurrence hypothesis: suppose the conditions are true up to j . Let $l \in S_m^{(j+1)}$. Since

$$S_R^{(j+1)} = S_r \cup \left(\bigcup_{j'=1}^j S_m^{(j')} \right),$$

we have

$$\begin{aligned}
y_l^* &= \tilde{\varphi}_{0l}^{(j+1)} + \sum_{k \in S_R^{(j+1)}} \tilde{\varphi}_{kl}^{(j+1)} y_k + \tilde{e}_l^* \\
&= \tilde{\varphi}_{0l}^{(j+1)} + \sum_{k \in S_r} \tilde{\varphi}_{kl}^{(j+1)} y_k + \sum_{j'=1}^j \sum_{l' \in S_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} y_{l'}^* + \tilde{e}_l^* \\
&= \tilde{\varphi}_{0l}^{(j+1)} + \sum_{k \in S_r} \tilde{\varphi}_{kl}^{(j+1)} y_k + \sum_{j'=1}^j \sum_{l' \in S_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \left[\varphi_{0l'}^{(j')} + \sum_{k \in S_r} \varphi_{kl'}^{(j')} y_k + e_{l'}^* \right] + \tilde{e}_l^*
\end{aligned}$$

where the last equality comes from the recurrence hypothesis. Reorganizing, we obtain

$$\begin{aligned}
y_l^* &= \tilde{\varphi}_{0l}^{(j+1)} + \sum_{j'=1}^j \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \varphi_{0l'}^{(j')} + \sum_{k \in s_r} \tilde{\varphi}_{kl}^{(j+1)} y_k + \sum_{j'=1}^j \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \left[\sum_{k \in s_r} \varphi_{kl'}^{(j')} y_k \right] + \tilde{e}_l^* \\
&\quad + \sum_{j'=1}^j \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} e_{l'}^* \\
&= \left\{ \tilde{\varphi}_{0l}^{(j+1)} + \sum_{j'=1}^j \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \varphi_{0l'}^{(j')} \right\} + \sum_{k \in s_r} \left[\tilde{\varphi}_{kl}^{(j+1)} + \sum_{j'=1}^j \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} \varphi_{kl'}^{(j')} \right] y_k \\
&\quad + \left\{ \tilde{e}_l^* + \sum_{j'=1}^j \sum_{l' \in s_m^{(j')}} \tilde{\varphi}_{l'l}^{(j)} e_{l'}^* \right\} \\
&= \varphi_{0l}^{(j+1)} + \sum_{k \in s_r} \varphi_{kl}^{(j+1)} y_k + e_l^*
\end{aligned}$$

which satisfies the conditions (6.6), (6.7) and (6.8) for $j+1$. This concludes the proof by recurrence. ■

Additional remarks:

1. Rewriting the imputed value y_l^* in the form (6.4) for all imputation steps of a variable of interest y is called the *canonical form* of the imputation strategy for y . When an imputation strategy fulfils the conditions of the proposition, we say that *L-imputation* (for linear imputation) has been applied to the variable y .
2. The derivation of $\varphi_{0l}^{(j)}$ and $\varphi_{kl}^{(j)}$ is done only from $\tilde{\varphi}_{0l}^{(j)}$ and $\tilde{\varphi}_{kl}^{(j)}$ and thus do not depend on the y -values.
3. If none of the J imputation steps use previously imputed y -values, then the notilde phi's are simply equal to the tilde phi's: $\varphi_{0l}^{(j)} \equiv \tilde{\varphi}_{0l}^{(j)}$, $\varphi_{kl}^{(j)} \equiv \tilde{\varphi}_{kl}^{(j)}$, $e_l^* \equiv \tilde{e}_l^*$, for $l \in s_m^{(j)}$, $k \in s_r$, $1 \leq j \leq J$.
4. When previously y -imputed nonrespondents were used as contributors, the quantity e_l^* , unlike $\tilde{e}_l^*$, does not represent anymore a random residual picked from a single contributor.
5. The proposition above does not restrict to a single imputation step, but in fact to all imputation steps related to a Y variable. Indeed, if some, but not all, imputation steps pass the condition (6.3), at some point the recurrence will break down to provide a

linear form as in (6.4).

6. When previously imputed non-respondents were used as contributors, it is worth noting that the imputed value y_l^* at step j may be built from respondents outside the imputation classes in usage at step j . Indeed, while the imputation contributors directly involved (those in $s_R^{(j)}$) come from the same class as the unit l imputed, some of these imputation contributors have in fact been imputed in other steps which may have used other sets of imputation classes. So at the end, when the form (6.4) is calculated, it is quite possible that $\varphi_{kl} \neq 0$ for k and l belonging to different classes as defined by the set of classes of step j .

7. It was said that all imputation methods supported in G-EST methodology satisfy the form (6.3). Note however that there exist other imputation methods that do not satisfy (6.3). Such an example is

$$y_l^* = e^{\frac{1}{n_r} \sum_{k \in s_r} \log(y_k)}.$$

8. There are closed forms of $\tilde{\varphi}_{0l}^{(j)}$ and $\tilde{\varphi}_{0l}^{(j)}$ for the imputation methods supported by G-EST. However, unless all steps prohibit previously imputed nonrespondents to be used as imputation contributors, there are no elegant forms for $\varphi_{0l}^{(j)}$ and $\varphi_{0l}^{(j)}$. The recurrence formulas help to derive easily their numeric values but not their algebraic forms.
9. When previously imputed values are allowed as imputation contributors, it is a misconception to think the imputation method at step j to really be, say, strictly linear regression imputation or strictly donor imputation or strictly something else, because the previously imputed values bring their own imputation methods with them to mingle with the current imputation method. An example after the remarks will show this.

10. Up to now, we have used the superscript j in $\varphi_{0l}^{(j)}$ and $\varphi_{kl}^{(j)}$ to pinpoint the step. However notice that $\varphi_{0l}^{(j)}$ and $\varphi_{kl}^{(j)}$ are defined only for $l \in s_m^{(j)}$, and the sets $s_m^{(1)}, \dots, s_m^{(j)}$ are mutually exclusive. Therefore with no confusion, we can define the following quantities:

$$\begin{aligned} \varphi_{0l} &= \varphi_{0l}^{(j)}, & l \in s_m^{(j)}, & \quad 1 \leq j \leq J, \\ \varphi_{kl} &= \varphi_{kl}^{(j)}, & l \in s_m^{(j)}, & \quad 1 \leq j \leq J. \end{aligned}$$

Note that φ_{0l} and φ_{kl} are defined for all $l \in s_m$, because it is assumed that all nonresponse has been successfully treated by imputation.

Example:

Consider the two-step imputation strategy below:

1. At first step, impute y by ratio imputation, using x as the auxiliary variable.
2. At second step, impute y by mean imputation, allowing previously imputed data to be used in the mean derivation.

Mathematically, this amounts to:

Step 1:

$$y_l^* = x_l \left\{ \sum_{k \in s_r \cap E} y_k / \sum_{k \in s_r \cap E} x_k \right\},$$

where $E = \{k \in s : x_k \text{ is non missing}\}$, for $l \in s_m^{(1)} = s_m \cap E$.

Step 2:

$$y_l^* = \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \sum_{k \in s_r \cup s_m^{(1)}} y_{\cdot k},$$

where $\text{card}(B)$ is the number of elements in the set B , for any set B , and $l \in s_m - E$.

The two imputation methods above have the form (6.2). In particular, we see that the sum in the step 2 is done also over imputed y -values.

At step 2, we have:

$$\begin{aligned} y_l^* &= \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \sum_{k \in s_r \cup s_m^{(1)}} y_{\cdot k} = \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \left\{ \sum_{k \in s_r} y_k + \sum_{l' \in s_m^{(1)}} y_{l'}^* \right\} \\ &= \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \left\{ \sum_{k \in s_r} y_k + \sum_{l' \in s_m^{(1)}} x_{l'} \frac{\sum_{k \in s_r \cap E} y_k}{\sum_{k \in s_r \cap E} x_k} \right\} \\ &= \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \sum_{k \in s_r} \left[1 + \frac{I_{E,k} \sum_{l' \in s_m^{(1)}} x_{l'}}{\sum_{k' \in s_r \cap E} x_{k'}} \right] y_k \end{aligned}$$

Thus the *canonical form* is

Step 1:

$$y_l^* = x_l \frac{\sum_{k \in s_r \cap E} y_k}{\sum_{k \in s_r \cap E} x_k},$$

where $E = \{k \in s : x_k \text{ is non missing}\}$, for $l \in s_m^{(1)} = s_m \cap E$.

Here

$$\varphi_{0l} \equiv 0, \quad l \in s_m \cap E,$$

$$\varphi_{kl} = \begin{cases} \frac{x_l}{\sum_{k \in s_r \cap E} x_k}, & l \in s_m \cap E, k \in s_r \cap E \\ 0, & l \in s_m \cap E, k \notin s_r \cap E \end{cases}$$

Step 2:

$$y_l^* = \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \sum_{k \in s_r} \left[1 + \frac{I_{E,k} x_{l'}}{\sum_{k' \in s_r \cap E} x_{k'}} \right] y_k$$

where $l \in s_m - E$ and $I_{E,k} = 1$ if $k \in E$, 0 otherwise.

Here

$$\varphi_{0l} \equiv 0, \quad l \in s_m - E,$$

$$\varphi_{kl} = \begin{cases} \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \left[1 + \frac{I_{E,k} x_{l'}}{\sum_{k' \in s_r \cap E} x_{k'}} \right], & l \in s_m - E, k \in s_r \\ 0, & l \in s_m - E, k \notin s_r \end{cases}$$

The canonical form, globally, is thus:

$$\varphi_{0l} \equiv 0, \quad l \in s_m,$$

$$\varphi_{kl} = \begin{cases} \frac{x_l}{\sum_{k \in s_r \cap E} x_k}, & l \in s_m \cap E, k \in s_r \cap E, \\ \frac{1}{\text{card}(s_r \cup s_m^{(1)})} \left[1 + \frac{I_{E,k} x_{l'}}{\sum_{k' \in s_r \cap E} x_{k'}} \right], & l \in s_m - E, k \in s_r \\ 0, & \text{otherwise.} \end{cases}$$

■

In G-EST, users will be able to use the traditional description (6.2) instead of the canonical form (6.4) for imputation steps, for two reasons:

1. The canonical description can become very complex to write down for the user, especially when there are many imputation steps.
2. In order to simplify parameter settings, the G-EST's interface will not accept a totally flexible syntax for describing the imputation methods.

6.6 Adjustment variable

Sometimes, an imputation method is not directly applied to the variable of interest y , but instead to the ratio of the variable of interest over another variable, say x_a . Once this ratio is imputed through the imputation method of choice, the imputed value of this ratio multiplied by the value of x_a for the nonrespondent is calculated and becomes the imputed value for y .

Mathematically, let $y_{ak} = y_k/x_{ak}$, for $k \in s_R \cap \{x_{ak} \neq 0\}$. The imputed value y_{al}^* , for $l \in s_m$, is derived as

$$y_{al}^* = f(s, l)$$

where f is the application of the imputation method. Then the imputed value y_l^* is derived as

$$y_l^* = x_{al} y_{al}^*.$$

While one may argue that we can enter x_a into the definition of the imputation method like

$$y_l^* = g(s, l)$$

where

$$g(s, l) = x_{al} f(s, l),$$

we will keep them separate because they induce a change in the underlying imputation model that will be discussed in chapter 7.

Note that in the discussion of the imputation methods below, we will assume there is an adjustment variable, although this may be a dummy one with $x_{ak} \equiv 1$ for all units.

Remark: The choice of the adjustment variable may change from one imputation step to the next, for the variable of interest y .

6.7 Family: Auxiliary value imputation

For AV imputation, the imputed value for a nonrespondent $l \in s_m$ is

$$y_l^* = x_{al} x_l$$

where x is a single auxiliary variable and x_a is an adjustment variable.

This imputation method satisfies the linearity (6.2) with

$$\tilde{\varphi}_{0l} = x_{al} x_l ,$$

$$\tilde{\varphi}_{kl} \equiv 0 .$$

Moreover it also satisfies (6.4) with

$$\varphi_{0l} = x_{al} x_l ,$$

$$\varphi_{kl} \equiv 0$$

because it does not depend on imputation contributors. This is a very peculiar family of imputation methods and the only one in G-EST free of imputation contributors.

Remark: In the current state of the methodology and of the implementation in G-Est, the AV imputation can only be deterministic (section 6.4).

Example 1: We are interested in imputing $y_{t,l}$, the value of variable y at time t for unit l . The imputed value is chosen to be $y_{t,l}^* = y_{t-1,l}$, where $y_{t-1,l}$ is the value of variable y at time $t-1$ for unit l . In this example, the adjustment variable is $x_{a,l} = 1$ and the auxiliary variable is $x_l \equiv y_{t-1,l}$. The current value $y_{t,l}$ is thus simply imputed by the previous value $y_{t-1,l}$.

Example 2: As in example 1, we are still interested in imputing $y_{t,l}$. The imputed value is now chosen to be $y_{t,l}^* = y_{t-1,l} a_{t,l} / a_{t-1,l}$, where $a_{t,l}$ and $a_{t-1,l}$ are values of an auxiliary variable a for unit l at time t and $t-1$, respectively. In this example, the adjustment variable is $x_{a,l} = a_{t,l}$ and the auxiliary variable is $x_l = y_{t-1,l} / a_{t-1,l}$. Here, the current ratio $y_{t,l} / a_{t,l}$ is imputed by the previous ratio $y_{t-1,l} / a_{t-1,l}$.

6.8 Family: Linear regression imputation

In the case of linear regression (LR) imputation, the imputed value for a nonrespondent l in the imputation class c is

$$y_l^* = x_{a,l} \mathbf{x}_l' \hat{\boldsymbol{\beta}}_c + \tilde{e}_l^*$$

where

$$\hat{\boldsymbol{\beta}}_c = \left(\sum_{k \in s_{R,c}} \frac{\omega_k^{\text{IMP}}}{v_k} \mathbf{x}_k \mathbf{x}_k' \right)^{-1} \sum_{k \in s_{R,c}} \frac{\omega_k^{\text{IMP}}}{v_k} \mathbf{x}_k (y_{\bullet k} / x_{a,k}),$$

$s_{R,c}$ is the set of imputation contributors belonging to the imputation class c , ω_k^{IMP} is the imputation weight for the contributor k , v_k is a variable for contributor k associated with the variance of the underlying model error (see section 7.5), $\mathbf{x}_k$ is a vector of auxiliary variables called *regressors* (which may include an intercept) and $\tilde{e}_l^*$ is either a random residual (if the method is stochastic, that is, $I_{\text{RESID},l} = 1$) or systematically zero (if the method is deterministic, that is, $I_{\text{RESID},l} = 0$).

When there is no adjustment variable x_a , no variable for the structure of variance v and all the imputation weights w_k^{IMP} are equal, the beta formula reduces to:

$$\hat{\boldsymbol{\beta}}_c = \left(\sum_{k \in s_{R,c}} \mathbf{x}_k \mathbf{x}_k' \right)^{-1} \sum_{k \in s_{R,c}} \mathbf{x}_k y_{\bullet k},$$

The imputation method, in its general form, is linear with

$$\tilde{\varphi}_{0l} \equiv 0,$$

$$\tilde{\varphi}_{kl} = x_{a,l} \left[\mathbf{x}_l' \left(\sum_{k' \in s_{R,c}} \frac{\omega_{k'}^{\text{IMP}}}{v_{k'}} \mathbf{x}_{k'} \mathbf{x}_{k'}' \right)^{-1} \mathbf{x}_k \right] \frac{\omega_k^{\text{IMP}}}{x_{a,k} v_k}, \quad k \in s_{R,c}, \quad l \in s_{m,c}.$$

where $s_{R,c}$ is the list of imputation contributors belonging to the imputation class c , c being one of the collection of imputation classes being in usage at this imputation step.

The LR family contains two well-used cases of imputation: the mean imputation and the ratio imputation.

6.8.1 Mean imputation

The mean imputation consists of imputing a recipient by the mean calculated from the imputation contributors of the same class:

$$y_l^* = \frac{\sum_{k \in s_{R,c}} y_{\bullet k}}{\text{card}(s_{R,c})}, \quad l \in s_{m,c}$$

which is LR imputation when the adjustment variable is equal to 1 for all units, there is no independent variable and there is an intercept.

6.8.2 Ratio imputation

The ratio imputation consists of imputing a recipient by its value of some variable x multiplied by the ratio of y over x calculated from the imputation contributors of the same class:

$$y_l^* = x_l \frac{\sum_{k \in s_{R,c}} y_{\bullet k}}{\sum_{k \in s_{R,c}} x_k}, \quad l \in s_{m,c}.$$

This is LR imputation where the adjustment variable is equal to 1 for all units, there is no intercept, the vector $\mathbf{x}_k$ consists of only the single value x_k and the variance structure v_k is x_k .

6.9 Family: Donor imputation

With donor imputation, the imputed value for a nonrespondent l is

$$y_l^* = x_{a,l} \frac{y_{d(l)}}{x_{a,d(l)}}$$

where $d(l)$ is the donor for the nonrespondent l taken among the imputation contributors.

If there is no adjustment variable x_a , the imputed value is simply

$$y_l^* = y_{d(l)}.$$

The imputation method, in its general form, is linear with

$$\tilde{\varphi}_{0l} = 0, \quad l \in s_m,$$

$$\tilde{\varphi}_{kl} = \begin{cases} x_{a,l} & , \quad l \in s_m, k = d(l) \\ x_{a,k} & \\ 0, & \text{otherwise} \end{cases}.$$

The choice of the donor $d(l)$ may be purely random or based on minimizing a distance between the nonrespondent and the imputation contributors. The first case is called random donor imputation (or hot deck). The second case is nearest-neighbour imputation (or nearest-donor imputation). In fact, a hybrid situation can even occur: nearest-neighbour imputation is attempted, but there is more than one donor at minimum distance for a nonrespondent, so one is randomly selected among the best ones. Note that in the context of donor imputation, a nonrespondent is usually called a recipient and an imputation contributor is usually called a potential donor.

The following is assumed: in the case of nearest-donor imputation, there exist one or more numeric auxiliary variables upon which the distance is based and these auxiliary variables are called the matching fields, while in the case of random donor imputation there is no matching field.

The exact form of the distance as a function of the matching fields in nearest-donor imputation is not taken into account in the current methodology, but the matching fields are.

Note also that in donor imputation, post-imputation edit rules can have been applied, which are conditions to be satisfied before accepting a donor to impute a recipient (for example of an edit rule: REVENUE1+REVENUE2+REVENUE3 is equal or almost equal to TOTALREVENUE). G-Est does not directly take account of these post-imputation edit rules, but it does indirectly since they influenced the choice of donors and the ID of the donor is taken into account in G-Est.

7 Imputation model

7.1 Imputation model approach

In G-EST, an imputation model approach to variance estimation is taken. The basic methodology is described in Särndal (1992) and was then developed further, among others, by Deville and Särndal (1994) for DLR imputation, Brick, Kalton and Kim (2004) and Beaumont and Bocci (2009) for donor imputation, Beaumont, Haziza and Bocci (2011) for AV imputation, and Beaumont and Bissonnette (2011) for composite imputation. In this approach, inference is made with respect to the joint distribution induced by the imputation model $\mathbb{m}$, the sampling design $\mathbb{p}$, the nonresponse mechanism $\mathbb{q}$ and the imputation mechanism $\star$, if any. The nonresponse

mechanism is the random mechanism that generates missing values. The imputation model is a set of assumptions about the variable requiring imputation (the y-variable). Greater detail about imputation models considered in G_EST is given in the next section.

7.2 General imputation model in G-EST

In G-EST, we consider the following general imputation model:

$$E_{\text{mm}}(y_k | \mathbf{x}_k^{\text{obs}}) = x_{a,k} \mu_{a,k} = \mu_k, \quad (7.1)$$

$$V_{\text{mm}}(y_k | \mathbf{x}_k^{\text{obs}}) = x_{a,k}^2 \sigma_{a,k}^2 = \sigma_k^2, \quad (7.2)$$

$$\text{cov}_{\text{mm}}(y_k, y_l | \mathbf{x}_k^{\text{obs}}, \mathbf{x}_l^{\text{obs}}) = 0, \quad (7.3)$$

for $k \neq l$, where $\mu_{a,k}$ and $\sigma_{a,k}^2$ are parametric or nonparametric functions of observed auxiliary variables for unit k . The full vector of observed auxiliary variables for unit k , which also includes the adjustment variable $x_{a,k}$ and imputation class indicators, is denoted by $\mathbf{x}_k^{\text{obs}}$. The above imputation model can also be rewritten as

$$E_{\text{mm}}(y_{a,k} | \mathbf{x}_k^{\text{obs}}) = \mu_{a,k},$$

$$V_{\text{mm}}(y_{a,k} | \mathbf{x}_k^{\text{obs}}) = \sigma_{a,k}^2,$$

$$\text{cov}_{\text{mm}}(y_{a,k}, y_{a,l} | \mathbf{x}_k^{\text{obs}}, \mathbf{x}_l^{\text{obs}}) = 0,$$

where $y_{a,k} = y_k / x_{a,k}$. This leads to a more direct interpretation of $\mu_{a,k}$ and $\sigma_{a,k}^2$. Since we will always condition on $\mathbf{x}_k^{\text{obs}}$ in the rest of this document, we remove this notation for simplicity, so that, for instance, $E_{\text{mm}}(y_k | \mathbf{x}_k^{\text{obs}})$ will be subsequently written $E_{\text{mm}}(y_k)$.

Note that the conditional uncorrelation of the imputation units in (7.3) under the imputation model is said to hold even between imputation units coming from the same sampling unit (in the general framework where a sampling unit may be divided in many imputation units).

In addition to the above imputation model assumptions, we make the following assumption:

$$F(\mathbf{Y} | s, s_r, \mathbf{X}^{\text{obs}}, \tilde{\mathbf{Z}}, \mathbf{D}) = F(\mathbf{Y} | \mathbf{X}^{\text{obs}})$$

where $F(\cdot)$ denotes the multivariate distribution function, $\mathbf{Y}$ and $\mathbf{D}$ are N -element vectors containing respectively y_k and d_k as their k^{th} element, $\mathbf{X}^{\text{obs}}$ is a N -row matrix containing $\mathbf{x}_k^{\text{obs}}$ in its k^{th} row and $\tilde{\mathbf{Z}}$ is a N -row matrix of design and calibration information, which implicitly or

explicitly contains information about the selection probabilities and the joint selection probabilities for $k, l \in U$. The distribution $F(\mathbf{Y}|\mathbf{X}^{\text{obs}})$ is more explicitly specified in the description of the imputation model at the beginning of this section. The above assumption allows G-EST to treat the sample, the set of respondents, the domain information, and the design and calibration information as being fixed when taking model expectations and model variances. A careful choice of the auxiliary variables is necessary at the imputation stage to make this assumption reasonable. For instance, the design and domain variables should be considered as potential auxiliary variables.

7.3 Link with imputation steps

The vector $\mathbf{x}_k^{\text{obs}}$ of observed auxiliary variables may not be always the same from a unit to the next. Usually this vector will determine which imputation method is used to impute a nonrespondent $l \in s_m$ or which imputation method would have been used for a respondent $k \in s_r$, had this respondent been actually a nonrespondent. Let us assume that imputation step j is assigned to a given unit k (a respondent or a nonrespondent) based on the vector $\mathbf{x}_k^{\text{obs}}$. Since the general imputation model defined at the beginning of section 7.1 conditions on the observed auxiliary variables, the estimated model mean and variance for this unit k are those obtained from imputation step j , which depends mainly on the family of imputation methods used. Sections 7.4, 7.5 and 7.6 give fully parametric estimation of $\mu_{a,k}$ and $\sigma_{a,k}^2$ for AV imputation, LR imputation and donor imputation. In section 7.7, semi-parametric estimation of $\mu_{a,k}$ and $\sigma_{a,k}^2$ is described.

Once estimators $\hat{\mu}_{a,k}$ and $\hat{\sigma}_{a,k}^2$ have been obtained, μ_k and σ_k^2 are then estimated by $\hat{\mu}_k = x_{a,k} \hat{\mu}_{a,k}$ and $\hat{\sigma}_k^2 = x_{a,k} \hat{\sigma}_{a,k}^2$ respectively.

Remark 1: The imputation model is not specified explicitly in G-EST by the user. Instead, it is G-EST that will derive the imputation model for each unit from the imputation step assigned to it.

Remark 2: In G-EST, the user must provide information on which imputation step was used to impute each nonrespondent. The user may provide this information for the respondents as well or let the System determine it. In the latter case, the order of the imputation steps specified in G-EST is important (in addition to be useful to determine imputation contributors if nonrespondents imputed at a previous step can be used; cf. § 6.2). First, G-EST checks if the first imputation step could be used for imputation of respondent k given the available auxiliary variables for this respondent. If the answer is positive, the first step is assigned to respondent k . Otherwise, G-EST checks if the second imputation step could be used and assigns it respondent k if it so. This procedure continues until a step has been assigned to respondent k .

Remark 3: One of the constraints to assign an imputation step to a respondent is to ensure there are enough units to build a stable model. These units are called the *modeling contributors*. They are very much related to the imputation contributors (section 6.2), except for two differences:

1. Only respondents can be used as modeling contributors;
2. Modeling contributors exist for AV imputation, while imputation contributors do not.

G-EST will use the imputation contributors of the step that are also respondents as the modeling contributors. In the case of AV, it will use the respondents that have no missing values for the auxiliary variable used as a substitution (and no missing values for the adjustment variable, if any).

Remark 4: To decide if the number of modeling contributors is large enough, the System uses imputation classes, mimicking what was done during imputation. However the imputation process may have used as imputation contributors nonrespondents imputed in previous steps, thus increasing the number of imputation contributors. On the other hand, modeling contributors cannot be nonrespondents. Therefore a class that was seen large enough for imputation may be too small for modeling. To palliate that, modeling could use an alternate collection of imputation classes, usually some collapsing of the true imputation classes, specified by the user for the imputation step. G-EST will first see if the class of the original collection of classes is large enough. If it is, modeling will be done using only the contributors of this class. If however this original class is too poor in modeling contributors, before G-EST checks the next imputation step, it will check again with the same step but using the alternate imputation class. Now if the number of modeling contributors is large enough, the modeling will be done using all modeling contributors from this (usually larger) alternate class. If however even in this second class there are still not enough contributors, the imputation step will be deemed unsuitable for the respondent k and G-EST will go to the next imputation step.

Remark 5: A respondent can be a modeling contributor for many imputation steps. However a unit, whether respondent or nonrespondent, can be assigned to only one imputation step, that is, it can be modeled and its values of μ_k and σ_k^2 can be estimated only from a single imputation model. Typically, if a respondent is a modeling contributor for some imputation step and another imputation step uses only a subset of the auxiliary variables of the first step, then the respondent will also be a modeling contributor for the second imputation step.

7.4 Parametric estimation: Auxiliary value imputation

For AV imputation, the parametric model for $\mu_{a,k}$ and $\sigma_{a,k}^2$ is $\mu_{a,k} = x_k$ and $\sigma_{a,k}^2 = \sigma_c^2 v_k$, where c is the imputation class of the unit k , v_k is a variable specified by the user that is associated with the error variance of the imputation model and σ_c^2 is an unknown model parameter for

imputation class c . The model mean $\mu_{a,k}$ and unknown model variance $\sigma_{a,k}^2$ are estimated by the model-unbiased estimators

$$\hat{\mu}_{a,k} = x_k$$

and

$$\hat{\sigma}_{a,k}^2 = \hat{\sigma}_c^2 v_k ,$$

where

$$\hat{\sigma}_c^2 = \sum_{k \in s_{\text{MOD},c}} \omega_k^{\text{MOD}} (y_{a,k} - \hat{\mu}_{a,k})^2 / \sum_{k \in s_{\text{MOD},c}} \omega_k^{\text{MOD}} v_k ,$$

$s_{\text{MOD},c}$ is the set of modeling contributors in class c and ω_k^{MOD} is the modeling weight. This modeling weight is determined by the imputation weights, except that its value is set to 0 for nonrespondents.

Note that, although AV imputation does not use imputation classes for filling in the missing values during the imputation process, the specification of imputation classes may nevertheless remain useful for the estimation of $\sigma_{a,k}^2$ or the semi-parametric estimation of $\mu_{a,k}$ (see section 7.7).

7.5 Parametric estimation: Linear regression imputation

With (LR) linear regression imputation, either with or without a random residual, the parametric model for $\mu_{a,k}$ and $\sigma_{a,k}^2$ is $\mu_{a,k} = \mathbf{x}'_k \boldsymbol{\beta}_c$ and $\sigma_{a,k}^2 = \sigma_c^2 v_k$, where c is the imputation class of unit k , $\boldsymbol{\beta}_c$ is a vector unknown model parameters for imputation class c , and σ_c^2 and v_k are defined as in section 7.4. The unknown model mean $\mu_{a,k}$ is estimated by the model-unbiased estimator

$$\hat{\mu}_{a,k} = \mathbf{x}'_k \hat{\boldsymbol{\beta}}_c$$

with $\hat{\boldsymbol{\beta}}_c$ given in section 6.8. As in AV imputation, the unknown model variance $\sigma_{a,k}^2$ is estimated by

$$\hat{\sigma}_{a,k}^2 = \hat{\sigma}_c^2 v_k ,$$

where

$$\hat{\sigma}_c^2 = \sum_{k \in s_{\text{MOD},c}} \omega_k^{\text{MOD}} (y_{a,k} - \hat{\mu}_{a,k})^2 / \sum_{k \in s_{\text{MOD},c}} \omega_k^{\text{MOD}} v_k .$$

7.6 Parametric estimation: Donor imputation

For donor imputation, the parametric linear model $\mu_{a,k} = \mathbf{x}'_k \boldsymbol{\beta}_c$ and $\sigma_{a,k}^2 = \sigma_c^2 v_k$ may again be appropriate as in linear regression imputation. The unknown model mean $\mu_{a,k}$ and model variance $\sigma_{a,k}^2$ are thus estimated as in section 7.5 for linear regression imputation.

When there are quantitative auxiliary variables, nearest-neighbour imputation is a common donor imputation method. The method can be viewed as being a nonparametric imputation method since the imputed estimator is approximately model-unbiased without requiring specifying the form of $\mu_{a,k}$ and $\sigma_{a,k}^2$ (see Beaumont and Bocci, 2009). One may thus prefer to consider a semi-parametric approach of estimating $\mu_{a,k}$ and $\sigma_{a,k}^2$ (see section 7.7) for nearest-neighbour imputation.

7.7 Semi-parametric estimation

Sometimes it may be suspected that a parametric imputation model for $\mu_{a,k}$ does not hold. G-EST offers a semi-parametric alternative, which may be particularly useful for nearest-neighbour imputation. First, $\mu_{a,k}$ is estimated parametrically as in sections 7.4, 7.5 or 7.6, depending on which family of imputation methods is used. Let us denote this parametric estimator by $\hat{\mu}_{a,k}^{\text{PAR}}$. Then $\mu_{a,k}$ is estimated using the nonparametric SAS procedure TPSPLINE, which performs smoothing splines based on penalized least-squares estimation (SAS Institute Inc., 2004). The estimated model mean for unit k in class c is the predicted value from this nonparametric regression between $y_{a,k}$ and $\hat{\mu}_{a,k}^{\text{PAR}}$. Note that respondents with $\omega_k^{\text{MOD}} = 0$ are excluded from computations. Finally, the resulting estimated model means are winsorized to avoid extreme predictions for the nonrespondents that could be unreliable. The smallest and largest estimated means among the respondents serve as the lower and upper winsorization thresholds respectively. The final semi-parametric estimate of $\mu_{a,k}$ is denoted by $\hat{\mu}_{a,k}$. It may happen that the procedure TPSPLINE fails to provide predicted means. When this occurs, $\mu_{a,k}$ is estimated parametrically.

There are two semi-parametric (or nonparametric) approaches of estimating $\sigma_{a,k}^2$. In the first method, $(y_{a,k} - \hat{\mu}_{a,k})^2$ is modeled as a function of $\hat{\mu}_{a,k}^{\text{PAR}}$ using the procedure TPSPLINE. The estimated model variance for unit k in class c is the predicted value from this nonparametric regression. Note again that respondents with $\omega_k^{\text{MOD}} = 0$ are excluded from the computations. This estimated model variance is then winsorized to avoid extreme predictions. The smallest and largest positive estimated variances among the respondents are the lower and upper winsorization thresholds respectively. The final semi-parametric estimate of $\sigma_{a,k}^2$ is denoted by $\hat{\sigma}_{a,k}^2$. It may happen again that the procedure TPSPLINE fails to provide predicted values. When

this occurs, $\sigma_{a,k}^2$ is estimated using an alternative semi-parametric approach described in the next paragraph.

A second semi-parametric (or nonparametric) approach available in G-EST estimates $\sigma_{a,k}^2$ by $\hat{\sigma}_{a,k}^2 = (y_{a,k} - \hat{\mu}_{a,k})^2$ for the respondent k . For the nonrespondent, $\sigma_{a,l}^2$ is estimated by

$$\hat{\sigma}_{a,l}^2 = \sum_{k \in D(l)} (y_{a,k} - \hat{\mu}_{a,k})^2$$

where $D(l)$ is the list of modeling contributors that share the same minimal distance with the nonrespondent l in the same imputation class. The distance between two units k and l is calculated as $|\hat{\mu}_{a,k}^{\text{PAR}} - \hat{\mu}_{a,l}^{\text{PAR}}|$. As usual, the modeling contributors are those respondents with $\omega_k^{\text{MOD}} > 0$.

It is to be noted that semi-parametric estimation is not performed when there is no auxiliary variable in the model (i.e. the model contains only an intercept), or when there are less than $\max(10, p+1)$ respondents in a class, where p is the number of parameters (number of auxiliary variables plus 1 if the model includes an intercept). In those cases, parametric estimation is performed.

8 Population parameters and their estimators

8.1 Population parameters

The two main population parameters supported by G-EST methodology are

the *generalized total*

$$T = \sum_{j \in U_{\varepsilon}} x_{\varepsilon,j} y_{\varepsilon,j} \quad (8.1)$$

and the *generalized ratio*

$$R = \frac{\sum_{j \in U_{\varepsilon}} x_{\varepsilon,j} y_{\varepsilon,j}}{\sum_{j \in U_{\varepsilon}} z_{\varepsilon,j}} \equiv \frac{T}{\sum_{j \in U_{\varepsilon}} z_{\varepsilon,j}} \quad (8.2)$$

where variables x and z are fully responded in the sample while the variable y may have nonresponse in the sample but this nonresponse is treated by linear imputation as described in section 6.5.

Please note that variables x and z above could be replaced without loss of generality by functions $g(\mathbf{x})$ and $h(\mathbf{z})$, where g and h are any known real-valued functions and $\mathbf{x}$ and $\mathbf{z}$ are any vectors of variables fully responded. However, the variable y cannot typically be replaced by some function $f(\mathbf{y})$ where each variable of the vector $\mathbf{y}$ was L-imputed, because the properties of linear imputation would not hold for the derived variable $y = f(\mathbf{y})$.

The generalized total and ratio give way to more usual population parameters as special cases.

Indeed, if we replace $x_{\varepsilon,j}$ by the domain indicator for a domain d , we get from (8.1) the *domain total*:

$$T_{yd} = \sum_{j \in U_{\varepsilon}} I_{dj} y_{\varepsilon,j} \quad (8.3)$$

where I_{dj} is equal to 1 if the estimation unit j belongs to the domain d and is equal to 0 otherwise. It is assumed here that the domain information is not imputed in the sample.

Also, if we replace both $x_{\varepsilon,j}$ and $z_{\varepsilon,j}$ by the domain inclusion for a domain d , we get from (8.2) the *domain mean*:

$$M_{yd} = \frac{\sum_{j \in U_{\varepsilon}} I_{dj} y_{\varepsilon,j}}{\sum_{j \in U_{\varepsilon}} I_{dj}} \equiv \frac{T_{yd}}{\sum_{j \in U_{\varepsilon}} I_{dj}} . \quad (8.4)$$

Finally, if we replace $x_{\varepsilon,j}$ by the domain inclusion for a domain d and $z_{\varepsilon,j}$ by the same domain inclusion multiplied by $z_{\varepsilon,j}$, we get from (8.2) the *domain ratio*:

$$R_{ydzd} = \frac{\sum_{j \in U_{\varepsilon}} I_{dj} y_{\varepsilon,j}}{\sum_{j \in U_{\varepsilon}} I_{dj} z_{\varepsilon,j}} \equiv \frac{T_{yd}}{\sum_{j \in U_{\varepsilon}} I_{dj} z_{\varepsilon,j}} . \quad (8.5)$$

All the population parameters above have been expressed in terms of estimation units; this makes sense since the domain variables are at this level. However, it will sometimes be convenient to write them at one of the two other unit levels. Some additional notation is required for that.

Using the allocation factors of (4.1), these population parameters become

$$T = \sum_{j \in U_{\varepsilon}} x_{\varepsilon,j} y_{\varepsilon,j} = \sum_{k \in U_j} \tilde{x}_{j,yk} y_{j,k} = \sum_{i \in U_s} \tilde{y}_{s,xi} , \quad (8.6)$$

$$R = \frac{\sum_{j \in U_{\varepsilon}} x_{\varepsilon,j} y_{\varepsilon,j}}{\sum_{j \in U_{\varepsilon}} z_{\varepsilon,j}} = \frac{\sum_{k \in U_j} \tilde{x}_{j,yk} y_{j,k}}{\sum_{k \in U_j} z_{j,k}} = \frac{\sum_{i \in U_s} \tilde{y}_{s,xi}}{\sum_{i \in U_s} z_{s,i}} , \quad (8.7)$$

$$T_{yd} = \sum_{j \in U_{\mathcal{E}}} I_{dj} y_{\mathcal{E},j} = \sum_{k \in U_j} \tilde{d}_{j,yk} y_{j,k} = \sum_{i \in U_{\mathcal{S}}} \tilde{y}_{\mathcal{S},di} \quad (8.8)$$

$$M_{yd} = \frac{\sum_{j \in U_{\mathcal{E}}} I_{dj} y_{\mathcal{E},j}}{\sum_{j \in U_{\mathcal{E}}} I_{dj}} = \frac{\sum_{k \in U_j} \tilde{d}_{j,yk} y_{j,k}}{\sum_{k \in U_j} d_{l,k}} = \frac{\sum_{k \in U_{\mathcal{S}}} \tilde{y}_{\mathcal{S},di}}{\sum_{i \in U_{\mathcal{S}}} d_{\mathcal{S},i}} \quad (8.9)$$

$$R_{y zd} = \frac{\sum_{j \in U_{\mathcal{E}}} I_{dj} y_{\mathcal{E},j}}{\sum_{j \in U_{\mathcal{E}}} I_{dj} z_{\mathcal{E},j}} = \frac{\sum_{k \in U_j} \tilde{d}_{j,yk} y_{j,k}}{\sum_{k \in U_j} \tilde{d}_{j,zk} z_{j,k}} = \frac{\sum_{k \in U_{\mathcal{S}}} \tilde{y}_{\mathcal{S},di}}{\sum_{i \in U_{\mathcal{S}}} \tilde{z}_{\mathcal{S},di}} \quad (8.10)$$

where

$$\begin{aligned} f_{yj} &= \frac{y_{E,j}}{y_{l,k}}, \quad f_{zj} = \frac{z_{E,j}}{z_{l,k}}, \quad j \in U_{E,k}, \\ \tilde{x}_{l,yk} &= \sum_{j \in U_{E,k}} x_{E,j} f_{yj}, \quad \tilde{d}_{l,yk} = \sum_{j \in U_{E,k}} I_{dj} f_{yj}, \quad \tilde{d}_{l,zk} = \sum_{j \in U_{E,k}} I_{dj} f_{zj}, \quad d_{l,k} \\ &= \sum_{j \in U_{E,k}} I_{dj}, \quad k \in U_l, \\ d_{\mathcal{S},i} &= \sum_{k \in U_{l,i}} d_{l,k}, \quad \tilde{y}_{\mathcal{S},xi} = \sum_{k \in U_{l,i}} \tilde{x}_{l,yk} y_{l,k}, \quad \tilde{y}_{\mathcal{S},di} = \sum_{k \in U_{l,i}} \tilde{d}_{l,yk} y_{l,k}, \quad \tilde{z}_{\mathcal{S},di} = \sum_{k \in U_{l,i}} \tilde{d}_{l,zk} z_{l,k}, \\ &\quad i \in U_{\mathcal{S}}. \end{aligned} \quad (8.11)$$

Finally note that when we want to talk in general about a population parameter and do not want to specify which one, we will use the notation θ as a generic population parameter. When more specification is required, θ will be replaced by the appropriate reference to T , R , T_{yd} , M_{yd} or $R_{y zd}$.

8.2 Non-traditional means as population parameters

The domain mean presented in (8.4) uses as the denominator the formula $\sum_{j \in U_E} I_{dj}$, which is the number of estimation units in the domain. This seems natural, since the domain variables are defined at the level of estimation units.

But sometimes another way to define the domain mean may arise. Instead of counting the number of estimation units that are in the domain, one may want to count the number of sampling units in the domain. Here a formal definition is required, because the value of the

domain variable may change from one estimation unit to the next within the same sampling unit. Therefore we say that the sampling unit is in the domain if at least one of its estimation units is in the domain⁷.

One has to be careful with such a definition of a domain, because some additivity property that held when counting estimation units does not hold anymore when counting sampling units. Let's look at the following example to help understand.

Consider a population where the sampling unit is the enterprise, the estimation unit is the location and the domain variable is the province. There are three domains of interest: QUEBEC, ONTARIO and the combination of QUEBEC and ONTARIO. A location has a strict value for the province: it is in Quebec, it is in Ontario or it is elsewhere; it cannot be in more than one province. On the other hand, the enterprise may have locations in more than one province.

If $N_{E,1}$ is the number of all locations in Quebec, $N_{E,2}$ the number of all locations in Ontario, and $N_{E,3}$ the number of all locations in Quebec or Ontario, then we have the relation

$$N_{E,3} = N_{E,1} + N_{E,2} .$$

However, if we decide to count enterprises with at least one location in the domain, then we can separate enterprises in four groups: the group having at least one location in Quebec but none in Ontario ($N_{S,A}$ enterprises, say), the group having at least one location in Ontario but none in Quebec ($N_{S,B}$ enterprises, say), the group having at least one location in Quebec and at least one location in Ontario ($N_{S,AB}$ enterprises, say) and the group having no location in Quebec nor Ontario.

Therefore the number of enterprises in the domain QUEBEC is $N_{S,1} = N_{S,A} + N_{S,AB}$, the number of enterprises in the domain ONTARIO is $N_{S,2} = N_{S,B} + N_{S,AB}$, and the number of enterprises on the domain QUEBEC+ONTARIO is $N_{S,3} = N_{S,A} + N_{S,B} + N_{S,AB}$. Clearly $N_{S,3} \neq N_{S,1} + N_{S,2}$ unless $N_{S,AB} = 0$, thus the number of enterprises in the domain QUEBEC+ONTARIO is not equal to the number of enterprises in the domain QUEBEC plus the number of enterprises in the domain ONTARIO.

Let's go further and consider the revenue as the variable of interest and we are interested to get the domain means for the three domains aforementioned, where the means are relative to sampling units (enterprises), instead of estimation units (locations). The user can define three new variables on the population file at the location level:

⁷ Some purists might say the sampling unit is *partly* in the domain, when some but not all its estimation units are in the domain.

$$d_{1j} = \begin{cases} 1 & \text{location } j \text{ is in Quebec} \\ 0 & \text{otherwise} \end{cases},$$

$$d_{2j} = \begin{cases} 1 & \text{location } j \text{ is in Ontario} \\ 0 & \text{otherwise} \end{cases},$$

$$d_{3j} = \begin{cases} 1 & \text{location } j \text{ is in Quebec or in Ontario} \\ 0 & \text{otherwise} \end{cases}.$$

Then the domain mean for Quebec is

$$M_1 = \frac{\sum_{j:d_{1j}=1} y_j}{N_{S,A} + N_{S,AB}},$$

the domain mean for Ontario is

$$M_2 = \frac{\sum_{j:d_{2j}=1} y_j}{N_{S,B} + N_{S,AB}}$$

and the domain mean for Quebec+Ontario is

$$M_3 = \frac{\sum_{j:d_{3j}=1} y_j}{N_{S,A} + N_{S,B} + N_{S,AB}}.$$

Since $d_{3j} = 1$ if and only if exactly only one of d_{1j} and d_{2j} is equal to one, we have

$$M_3 = \frac{\sum_{j:d_{1j}=1} y_j + \sum_{j:d_{2j}=1} y_j}{N_{S,A} + N_{S,B} + N_{S,AB}} = \frac{N_{S,A} + N_{S,AB}}{N_{S,A} + N_{S,B} + N_{S,AB}} M_1 + \frac{N_{S,B} + N_{S,AB}}{N_{S,A} + N_{S,B} + N_{S,AB}} M_2.$$

We note that the coefficients $(N_{S,A} + N_{S,AB})/(N_{S,A} + N_{S,B} + N_{S,AB})$ and $(N_{S,B} + N_{S,AB})/(N_{S,A} + N_{S,B} + N_{S,AB})$ are positive and their sum, $1 + N_{S,AB}/(N_{S,A} + N_{S,B} + N_{S,AB})$, exceeds 1 (unless $N_{S,AB} = 0$). Therefore the value of M_3 is not necessarily a value between M_1 and M_2 . In fact, when the value of $N_{S,AB}$ is quite large relative to $N_{S,A}$ or $N_{S,B}$, the value of M_3 may be greater than both values M_1 and M_2 , which may seem unnatural to uninformed users.

In order to accommodate the notation from section 8.1, we define

$$d_{s,i} = \begin{cases} 1 & \text{if } \sum_{k \in U_{I,i}} \sum_{j \in U_{E,k}} I_{dj} > 0 \\ 0 & \text{otherwise} \end{cases} = \max(I_{dj} : j \in U_{E,k}, k \in U_{I,i})$$

Mathematically, counting the sampling units with at least one estimation unit in the domain can be expressed as

$$\sum_{i \in U_S: \exists j \in U_{E,i}, I_{dj}=1} 1 .$$

For the remaining of this document, we will use the notation N_d to represent the number of units to count, and the user of G-Est will be given the choice for the definition of N_d :

$$N_d = \begin{cases} \sum_{j \in U_E} I_{dj} & \text{to count estimation units} \\ \sum_{i \in U_S: \exists j \in U_{E,i}, I_{dj}=1} 1 & \text{to count sampling units} \\ \sum_{i \in U_\gamma: \exists j \in U_{\gamma,i}, I_{dj}=1} 1 & \text{to count units at some intermediate level} \end{cases} \quad (8.12)$$

The third row of the definition of N_d above allows a broad generalization, because U_γ is considered some nested level between the sampling level and the estimation level.

In light of this definition, we can reexpress any domain mean as

$$M_{yd} = \frac{\sum_{j \in U_E} I_{dj} y_{E,j}}{N_d} \equiv \frac{T_{Yd}}{N_d} .$$

8.3 Estimators

The estimators are simply sample-weighted versions of the respective population parameters, where the weight w is called the estimation weight and can be the design weight or some adjusted weight (through calibration or other processes; see chapter 4).

It is assumed that the weight is at the sampling level, that is, the same among all imputation units and all estimation units belonging to the same sampling unit. Therefore, if $w_{S,i}$, $w_{I,k}$ and $w_{E,j}$ denote respectively the estimation weight for the sampling unit $i \in s_S$, the imputation unit $k \in s_I$ and the estimation unit $j \in s_E$, we have the following constraints:

$$\begin{aligned} w_{E,j} &\equiv w_{I,k}, & k \in s_I, j \in U_{E,k} , \\ w_{I,k} &\equiv w_{S,i}, & i \in s_S, k \in U_{I,i} . \end{aligned}$$

8.3.1 Estimators under full response

Under full response, the estimators are

$$\hat{T} = \sum_{j \in S_E} w_{E,j} x_{E,j} y_{E,j} = \sum_{k \in S_I} w_{I,k} \tilde{x}_{I,yk} y_{I,k} = \sum_{i \in S_S} w_{S,i} \tilde{y}_{S,xi} , \quad (8.13)$$

$$\hat{R} = \frac{\sum_{j \in S_E} w_{E,j} x_{E,j} y_{E,j}}{\sum_{j \in S_E} w_{E,j} z_{E,j}} = \frac{\sum_{k \in S_I} w_{I,k} \tilde{x}_{I,yk} y_{I,k}}{\sum_{k \in S_I} w_{I,k} z_{I,k}} = \frac{\sum_{i \in S_S} w_{S,i} \tilde{y}_{S,xi}}{\sum_{i \in S_S} w_{S,i} z_{S,i}} , \quad (8.14)$$

$$\hat{T}_{yd} = \sum_{j \in S_E} w_{E,j} I_{dj} y_{E,j} = \sum_{k \in S_I} w_{I,k} \tilde{d}_{I,yk} y_{I,k} = \sum_{i \in S_S} w_{S,i} \tilde{y}_{S,di} \quad (8.15)$$

$$\hat{M}_{yd} = \frac{\sum_{j \in S_E} w_{E,j} I_{dj} y_{E,j}}{\hat{N}_d} = \frac{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,yk} y_{I,k}}{\hat{N}_d} = \frac{\sum_{i \in S_S} w_{S,i} \tilde{y}_{S,di}}{\hat{N}_d} \quad (8.16)$$

$$\hat{R}_{yzd} = \frac{\sum_{j \in S_E} w_{E,j} I_{dj} y_{E,j}}{\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j}} = \frac{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,yk} y_{I,k}}{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,zk} z_{I,k}} = \frac{\sum_{i \in S_S} w_{S,i} \tilde{y}_{S,di}}{\sum_{i \in S_S} w_{S,i} \tilde{z}_{S,di}} \quad (8.17)$$

where $\hat{N}_d$ is the estimator of the count of units in the domain N_d (see (8.12)):

$$\hat{N}_d = \begin{cases} \sum_{j \in S_E} w_{E,j} I_{dj} & \text{to count estimation units} \\ \sum_{i \in U_S: \exists j \in U_{E,i}, I_{dj}=1} w_{S,i} & \text{to count sampling units} \\ \sum_{i \in U_\gamma: \exists j \in U_{\gamma,i}, I_{dj}=1} w_{\gamma,i} & \text{to count units at some intermediate level} \end{cases} \quad (8.18)$$

8.3.2 Estimators under nonresponse

In the general case where imputation may have occurred (only for the variable y), the estimators are

$$\hat{T}_\bullet = \sum_{j \in S_E} w_{E,j} x_{E,j} y_{E,\bullet j} = \sum_{k \in S_I} w_{I,k} \tilde{x}_{I,yk} y_{I,\bullet k} = \sum_{i \in S_S} w_{S,i} \tilde{y}_{S,\bullet xi} , \quad (8.19)$$

$$\hat{R}_\bullet = \frac{\sum_{j \in S_E} w_{E,j} x_{E,j} y_{E,\bullet j}}{\sum_{j \in S_E} w_{E,j} z_{E,j}} = \frac{\sum_{k \in S_I} w_{I,k} \tilde{x}_{I,yk} y_{I,\bullet k}}{\sum_{k \in S_I} w_{I,k} z_{I,k}} = \frac{\sum_{i \in S_S} w_{S,i} \tilde{y}_{S,\bullet xi}}{\sum_{i \in S_S} w_{S,i} z_{S,i}} , \quad (8.20)$$

$$\hat{T}_{\bullet yd} = \sum_{j \in S_E} w_{E,j} I_{dj} y_{E,\bullet j} = \sum_{k \in S_I} w_{I,k} \tilde{d}_{I,yk} y_{I,\bullet k} = \sum_{i \in S_S} w_{S,i} \tilde{y}_{S,\bullet di} , \quad (8.21)$$

$$\hat{M}_{\bullet yd} = \frac{\sum_{j \in S_E} w_{E,j} I_{dj} y_{E,\bullet j}}{\hat{N}_d} = \frac{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,yk} y_{I,\bullet k}}{\hat{N}_d} = \frac{\sum_{i \in S_S} w_{S,i} \tilde{y}_{S,\bullet di}}{\hat{N}_d} , \quad (8.22)$$

$$\hat{R}_{\bullet yzd} = \frac{\sum_{j \in S_E} w_{E,j} I_{dj} y_{E,\bullet j}}{\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j}} = \frac{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,yk} y_{I,\bullet k}}{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,zk} z_{I,k}} = \frac{\sum_{i \in S_S} w_{S,i} \tilde{y}_{S,\bullet di}}{\sum_{i \in S_S} w_{S,i} \tilde{z}_{S,\bullet di}} \quad (8.23)$$

where

$$y_{E,\bullet j} = f_{yk} y_{I,\bullet k} , \quad j \in U_{E,k} ,$$

$$\tilde{y}_{S,\bullet xi} = \sum_{k \in U_{I,i}} \tilde{x}_{I,yk} y_{I,\bullet k} , \quad \tilde{y}_{S,\bullet di} = \sum_{k \in U_{I,i}} \tilde{d}_{I,yk} y_{I,\bullet k} , \quad i \in S_S .$$

8.3.3 Canonical form of the estimators

Now, under the assumption that the imputation strategy of the variable y is linear (section 6.5), it means that

$$y_{I,l}^* = \varphi_{0l} + \sum_{k \in S_{I,r}} \varphi_{kl} y_{I,k} + e_{I,l}^* , \quad l \in S_{I,m} .$$

Therefore the estimators can be rewritten as linear combinations of y -values of respondents only:

$$\hat{T}_{\bullet} = W_{x0} + \sum_{k \in S_{I,r}} (w_{I,k} \tilde{x}_{I,yk} + W_{I,xk}) y_{I,k} + \sum_{l \in S_{I,m}} w_{I,l} \tilde{x}_{I,yl} e_{I,l}^* , \quad (8.24)$$

$$\hat{R}_{\bullet} = \frac{W_{x0} + \sum_{k \in S_{I,r}} (w_{I,k} \tilde{x}_{I,yk} + W_{I,xk}) y_{I,k} + \sum_{l \in S_{I,m}} w_{I,l} \tilde{x}_{I,yl} e_{I,l}^*}{\sum_{k \in S_I} w_{I,k} z_{I,k}} , \quad (8.25)$$

$$\hat{T}_{\bullet yd} = W_{d0} + \sum_{k \in S_{I,r}} (w_{I,k} \tilde{d}_{I,yk} + W_{I,dk}) y_{I,k} + \sum_{l \in S_{I,m}} w_{I,l} \tilde{d}_{I,yl} e_{I,l}^* , \quad (8.26)$$

$$\hat{M}_{\bullet yd} = \frac{W_{d0} + \sum_{k \in S_{I,r}} (w_{I,k} \tilde{d}_{I,yk} + W_{I,dk}) y_{I,k} + \sum_{l \in S_{I,m}} w_{I,l} \tilde{d}_{I,yl} e_{I,l}^*}{\hat{N}_d} , \quad (8.27)$$

$$\hat{R}_{\bullet yzd} = \frac{W_{d0} + \sum_{k \in S_{I,r}} (w_{I,k} \tilde{d}_{I,yk} + W_{I,dk}) y_{I,k} + \sum_{l \in S_{I,m}} w_{I,l} \tilde{d}_{I,yl} e_{I,l}^*}{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,zk} z_{I,k}} \quad (8.28)$$

where

$$\begin{aligned}
 W_{x0} &= \sum_{l \in S_{I,m}} w_{I,l} \tilde{x}_{I,yl} \varphi_{0l} , & W_{d0} &= \sum_{l \in S_{I,m}} w_{I,l} \tilde{d}_{I,yl} \varphi_{0l} , \\
 W_{I,xk} &= \sum_{l \in S_{I,m}} w_{I,l} \tilde{x}_{I,yl} \varphi_{kl} , & W_{I,dk} &= \sum_{l \in S_{I,m}} w_{I,l} \tilde{d}_{I,yl} \varphi_{kl} , & k \in S_{I,r} .
 \end{aligned} \tag{8.29}$$

Note that W_{x0} , W_{d0} , $W_{I,xk}$ and $W_{I,dk}$ do not depend on the y -values.

The forms for the estimators above are linear combinations of y -values from the respondents (except for the random terms $\tilde{E}_x = \sum_{l \in S_{I,m}} w_{I,l} \tilde{x}_{I,yl} e_{I,l}^*$ and $\tilde{E}_d = \sum_{l \in S_{I,m}} w_{I,l} \tilde{d}_{I,yl} e_{I,l}^*$), where the multiplicative and additive coefficients of the y -values do not depend on imputed variables (and thus will be considered fixed when conditioning under the imputation model). As θ is a generic notation for any population parameter, $\hat{\theta}$ will be used for the estimator in the case of full response and $\hat{\theta}_\bullet$ for the estimator in presence of imputation for the variable y . Note that $\hat{\theta}_\bullet$, $\hat{T}_\bullet$, $\hat{R}_\bullet$, $\hat{T}_{\bullet yd}$, $\hat{M}_{\bullet yd}$ and $\hat{R}_{\bullet yd}$ are sometimes called the imputed estimators, to differentiate them from the estimators where there is full response.

8.4 Estimation errors

8.4.1 Total error

The total error is defined as $\hat{\theta}_\bullet - \theta$. We consider the usual decomposition for the total error:

$$\hat{\theta}_\bullet - \theta = (\hat{\theta} - \theta) + (\hat{\theta}_\bullet - \hat{\theta}). \tag{8.30}$$

8.4.2 Sampling error

The first term in the right-hand-side of equation (8.30), $(\hat{\theta} - \theta)$, is called the sampling error. It will be useful to decompose this sampling error into sums over the respondents, the nonrespondents and the non-sampled units. However, remember that the notion of respondent versus nonrespondent ends at the imputation level, because a sampling unit can be composed of respondents and nonrespondents as well.

For the generalized total

Using (8.6) and (8.15), we obtain

$$\begin{aligned}
 \hat{T} - T &= \sum_{j \in S_{E,r}} (w_{E,j} - 1) x_{E,j} y_{E,j} + \sum_{j \in S_{E,m}} (w_{E,j} - 1) x_{E,j} y_{E,j} - \sum_{j \in U_E - S_E} x_{E,j} y_{E,j} \\
 &= \sum_{k \in S_{I,r}} (w_{I,k} - 1) \tilde{x}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} (w_{I,k} - 1) \tilde{x}_{I,yk} y_{I,k} - \sum_{k \in U_I - S_I} \tilde{x}_{I,yk} y_{I,k} \\
 &= \sum_{i \in S_S} (w_{S,i} - 1) \tilde{y}_{S,xi} - \sum_{i \in U_S - S_S} \tilde{y}_{S,xi}
 \end{aligned} \tag{8.31}$$

For the generalized ratio

Using (8.7) and (8.14), we get

$$\begin{aligned}
 \hat{R} - R &= \frac{\sum_{j \in S_{E,r}} (w_{E,j} - \rho_z) x_{E,j} y_{E,j} + \sum_{j \in S_{E,m}} (w_{E,j} - \rho_z) x_{E,j} y_{E,j} - \rho_z \sum_{j \in U_E - S_E} x_{E,j} y_{E,j}}{\sum_{j \in S_E} w_{E,j} z_{E,j}} \\
 &= \frac{\sum_{k \in S_{I,r}} (w_{I,k} - \rho_z) \tilde{x}_{I,k} y_{I,k} + \sum_{k \in S_{I,m}} (w_{I,k} - \rho_z) \tilde{x}_{I,k} y_{I,k} - \rho_z \sum_{k \in U_I - S_I} \tilde{x}_{I,k} y_{I,k}}{\sum_{j \in S_E} w_{E,j} z_{E,j}} \\
 &= \frac{\sum_{i \in S_S} (w_{S,i} - \rho_z) \tilde{y}_{S,xi} - \rho_z \sum_{i \in U_S - S_S} \tilde{y}_{S,xi}}{\sum_{j \in S_E} w_{E,j} z_{E,j}}
 \end{aligned} \tag{8.32}$$

where

$$\rho_z = \frac{\sum_{j \in S_E} w_{E,j} z_{E,j}}{\sum_{j \in U_E} z_{E,j}}.$$

For the domain total

Using (8.8) and (8.15), we get

$$\begin{aligned}
 \hat{T}_{yd} - T_{yd} &= \sum_{j \in S_{E,r}} (w_{E,j} - 1) I_{dj} y_{E,j} + \sum_{j \in S_{E,m}} (w_{E,j} - 1) I_{dj} y_{E,j} - \sum_{j \in U_E - S_E} I_{dj} y_{E,j} \\
 &= \sum_{k \in S_{I,r}} (w_{I,k} - 1) \tilde{d}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} (w_{I,k} - 1) \tilde{d}_{I,yk} y_{I,k} - \sum_{k \in U_I - S_I} \tilde{d}_{I,yk} y_{I,k}
 \end{aligned} \tag{8.33}$$

$$= \sum_{i \in S_S} (w_{S,i} - 1) \tilde{y}_{S,di} - \sum_{i \in U_S - S_S} \tilde{y}_{S,di}$$

For the domain mean

Using (8.9) and (8.16), we get

$$\begin{aligned} & \hat{M}_{yd} - M_{yd} \\ &= \frac{\sum_{j \in S_{E,r}} (w_{E,j} - \rho_d) I_{dj} y_{E,j} + \sum_{j \in S_{E,m}} (w_{E,j} - \rho_d) I_{dj} y_{E,j} - \rho_d \sum_{j \in U_E - S_E} I_{dj} y_{E,j}}{\hat{N}_d} \\ &= \frac{\sum_{k \in S_{I,r}} (w_{I,k} - \rho_d) \tilde{d}_{I,k} y_{I,k} + \sum_{k \in S_{I,m}} (w_{I,k} - \rho_d) \tilde{d}_{I,k} y_{I,k} - \rho_d \sum_{k \in U_I - S_I} \tilde{d}_{I,k} y_{I,k}}{\hat{N}_d} \\ &= \frac{\sum_{i \in S_S} (w_{S,i} - \rho_d) \tilde{y}_{S,di} - \rho_d \sum_{i \in U_S - S_S} \tilde{y}_{S,di}}{\hat{N}_d} \end{aligned} \quad (8.34)$$

where

$$\rho_d = \frac{\hat{N}_d}{N_d}.$$

For the domain ratio

Using (8.10) and (8.17), we get

$$\begin{aligned} & \hat{R}_{yzd} - R_{yzd} \\ &= \frac{\sum_{j \in S_{E,r}} (w_{E,j} - \rho_{zd}) I_{dj} y_{E,j} + \sum_{j \in S_{E,m}} (w_{E,j} - \rho_{zd}) I_{dj} y_{E,j} - \rho_{zd} \sum_{j \in U_E - S_E} I_{dj} y_{E,j}}{\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j}} \\ &= \frac{\sum_{k \in S_{I,r}} (w_{I,k} - \rho_{zd}) \tilde{d}_{I,k} y_{I,k} + \sum_{k \in S_{I,m}} (w_{I,k} - \rho_{zd}) \tilde{d}_{I,k} y_{I,k} - \rho_{zd} \sum_{k \in U_I - S_I} \tilde{d}_{I,k} y_{I,k}}{\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j}} \\ &= \frac{\sum_{i \in S_S} (w_{S,i} - \rho_{zd}) \tilde{y}_{S,xi} - \rho_{zd} \sum_{i \in U_S - S_S} \tilde{y}_{S,di}}{\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j}} \end{aligned} \quad (8.35)$$

where

$$\rho_{zd} = \frac{\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j}}{\sum_{j \in U_E} I_{dj} z_{E,j}}.$$

8.4.3 Nonresponse error

The second term in the right-hand side of the total error (8.30), $(\hat{\theta}_\bullet - \hat{\theta})$, is called the nonresponse error. Using equations from sections 8.3.1 and 8.3.3, we get expressions in terms of respondents and nonrespondents.

For the generalized total

$$\hat{T}_\bullet - \hat{T} = W_{x0} + \sum_{k \in S_{I,r}} W_{xk} y_{I,k} - \sum_{k \in S_{I,m}} w_{I,k} \tilde{x}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} w_{I,k} \tilde{x}_{I,yk} e_{I,k}^* . \quad (8.36)$$

For the generalized ratio

$$\hat{R}_\bullet - \hat{R} = \frac{W_{d0} + \sum_{k \in S_{I,r}} W_{dk} y_{I,k} - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} e_{I,k}^*}{\sum_{k \in S_I} w_{I,k} z_{I,k}} . \quad (8.37)$$

For the domain total

$$\hat{T}_{\bullet yd} - \hat{T}_{yd} = W_{d0} + \sum_{k \in S_{I,r}} W_{dk} y_{I,k} - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} e_{I,k}^* . \quad (8.38)$$

For the domain mean

$$\begin{aligned} & \hat{M}_{\bullet yd} - \hat{M}_{yd} \\ &= \frac{W_{d0} + \sum_{k \in S_{I,r}} W_{dk} y_{I,k} - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} e_{I,k}^*}{\hat{N}_d} . \end{aligned} \quad (8.39)$$

For the domain ratio

$$\begin{aligned} & \hat{R}_{\bullet yzd} - \hat{R}_{yzd} \\ &= \frac{W_{d0} + \sum_{k \in S_{I,r}} W_{dk} y_{I,k} - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} y_{I,k} + \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} e_{I,k}^*}{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,zk} z_{I,k}} \end{aligned} \quad (8.40)$$

9 Inference

9.1 Model assumptions

A basic assumption is that the estimator $\hat{\theta}_{\bullet}$ is unbiased or approximately unbiased. Under the imputation model approach to inference, this assumption means that the bias $E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \theta)$ is negligible (at least asymptotically) compared with the squared root of the variance $V_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \theta)$; this is denoted by

$$E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \theta) \approx 0. \quad (9.1)$$

Since the complete response estimator $\hat{\theta}$, either based on Horvitz-Thompson or calibrated weights, is exactly or at least approximately design-unbiased, which is written $E_{\mathbb{p}}(\hat{\theta} - \theta) \approx 0$, this assumption is thus equivalent to assuming that $E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \hat{\theta}) \approx 0$. This equivalence can be easily shown by using the standard decomposition of the total error, $\hat{\theta}_{\bullet} - \theta = (\hat{\theta} - \theta) - (\hat{\theta}_{\bullet} - \hat{\theta})$ (see (8.30)).

The first term on the right-hand side of this equation, $\hat{\theta} - \theta$, is usually called the sampling error while the second term, $\hat{\theta}_{\bullet} - \hat{\theta}$, is called the nonresponse error. A bias expression is given explicitly in section 9.2.

The above error decomposition can also be used to express the total mean squared error, $E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \theta)^2$, as

$$\begin{aligned} E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \theta)^2 &= E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta} - \theta)^2 + E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta}_{\bullet} - \hat{\theta})^2 + 2 E_{\mathbb{m}\mathbb{p}\mathbb{q}\star}(\hat{\theta} - \theta)(\hat{\theta}_{\bullet} - \hat{\theta}) \\ &\approx E_{\mathbb{m}} V_{\mathbb{p}}(\hat{\theta}) + E_{\mathbb{p}\mathbb{q}} E_{\mathbb{m}\star} [(\hat{\theta}_{\bullet} - \hat{\theta})^2 | s_I, s_{I,r}] \\ &\quad + 2 E_{\mathbb{p}\mathbb{q}} E_{\mathbb{m}\star} [(\hat{\theta} - \theta)(\hat{\theta}_{\bullet} - \hat{\theta}) | s_I, s_{I,r}]. \end{aligned}$$

The approximation in the previous equation is due to $E_{\mathbb{p}}(\hat{\theta} - \theta) \approx 0$. The first term of the right-hand side of the approximation, $V_{\text{SAM}} = E_{\mathbb{m}} V_{\mathbb{p}}(\hat{\theta})$, is the sampling variance, the second term,

$V_{NR} = E_{\mathbb{PQ}} E_{\mathbb{M}\star} [(\hat{\theta}_{\bullet} - \hat{\theta})^2 | s_I, s_{I,r}]$, is the nonresponse variance and the third term, $V_{MIX} = 2E_{\mathbb{PQ}} E_{\mathbb{M}\star} [(\hat{\theta} - \theta)(\hat{\theta}_{\bullet} - \hat{\theta}) | s_I, s_{I,r}]$, is a mixed component. The sum of the last two terms, $C_{NR} = V_{NR} + V_{MIX}$, is called the nonresponse component as it disappears in the absence of nonresponse. But on the other hand, in the case of a census, V_{SAM} and V_{MIX} disappear. So V_{MIX} is zero whenever there is a census or there is full response or both. Estimation of these three terms is described in sections 9.3, 9.4 and 9.5.

Note that the total mean squared error of $\hat{\theta}_{\bullet}$ can be called the total variance of $\hat{\theta}_{\bullet}$ when the unbiasedness assumption $E_{\mathbb{M}\mathbb{PQ}\star}(\hat{\theta}_{\bullet} - \theta) \approx 0$ holds because $E_{\mathbb{M}\mathbb{PQ}\star}(\hat{\theta}_{\bullet} - \theta)^2 \approx V_{\mathbb{M}\mathbb{PQ}\star}(\hat{\theta}_{\bullet} - \theta)$. It is also worth noting that the nonresponse component can be rewritten as

$$C_{NR} = E_{\mathbb{PQ}} E_{\mathbb{M}\star} [(\hat{\theta}_{\bullet} - \theta)^2 - (\hat{\theta} - \theta)^2 | s_I, s_{I,r}].$$

It is thus the average of the difference between the squared total error and the squared sampling error. Both the mixed component and the nonresponse component can be negative although this is less likely for the latter.

Now define

$$\tilde{\mu}_{S,i} = E_{\mathbb{M}}(y_{S,i} | \mathbf{Y}_r), \quad (9.2)$$

$$\tilde{\sigma}_{S,i}^2 = \text{var}_{\mathbb{M}}(y_{S,i} | \mathbf{Y}_r), \quad (9.3)$$

which are the expectation and variance of the variable of interest at the sampling level, under the imputation model and conditionally to the responded y -values.

Since $y_{S,i} = \sum_{k \in U_{I,i}} y_{I,k}$ and because of equations (7.1), (7.2) and (7.3) from section 7.2, one has

$$\tilde{\mu}_{S,i} = \sum_{k \in U_{I,i}} [r_{I,k} y_{I,k} + (1 - r_{I,k}) \mu_{I,k}]$$

and

$$\tilde{\sigma}_{S,i}^2 = \sum_{k \in U_{I,i}} (1 - r_{I,k}) \sigma_{I,k}^2$$

where $r_{I,k}$ is the response indicator for the imputation unit k . Note that these quantities, at the sampling level, are different from $\mu_{I,k}$ and $\sigma_{I,k}^2$, which are defined at the imputation level, in the following regard: the former are defined by conditioning also on the responded y -values, unlike the latter. This explains why $y_{I,k}$ appears in $\tilde{\mu}_{S,i}$ instead of $\mu_{I,k}$, when k is a respondent, and also why the sum in $\tilde{\sigma}_{S,i}^2$ is done only over the nonrespondents.

9.2 Bias

As pointed out in section 9.1, a basic assumption is that the imputed estimator $\hat{\theta}_\bullet$ is unbiased or approximatively unbiased, which is denoted by (9.1). We also pointed out that this assumption is equivalent to $E_{\mathbb{P}_{\mathbb{Q}}^*}(\hat{\theta}_\bullet - \hat{\theta}) \approx 0$, which can be rewritten as $E_{\mathbb{P}_{\mathbb{Q}}} E_{\mathbb{M}^*}[(\hat{\theta}_\bullet - \hat{\theta})|s_I, s_{I,r}] \approx 0$. Therefore the bias is negligible when the conditional bias $B_c = E_{\mathbb{M}^*}[(\hat{\theta}_\bullet - \hat{\theta})|s_I, s_{I,r}]$ is negligible. It is easy to show that this conditional bias is zero when AV or linear regression imputation is used along with a parametric model for $\mu_{a,k}$ because $E_{\mathbb{M}^*}[(y_{I,k} - y_{I,k}^*)|s_I, s_{I,r}] = 0$. For donor imputation, Beaumont and Bocci (2009) showed that this conditional bias is negligible under some weak conditions. However, the bias may not always be negligible, especially under composition imputation (that is, more than one imputation step), even when the differences between steps is only hierarchical imputation classes, or under AV imputation and linear regression imputation when a parametric model is not assumed for $\mu_{a,k}$.

To evaluate the magnitude of the bias, it is possible to estimate the conditional bias B_c . We will first find an expression for the nonresponse error $\hat{\theta}_\bullet - \hat{\theta}$. Then using this expression, the conditional bias will be obtained by taking the expectation $B_c = E_{\mathbb{M}^*}[(\hat{\theta}_\bullet - \hat{\theta})|s_I, s_{I,r}]$, which depends on the unknown model means μ_k . The estimated conditional bias $\hat{B}_c$ will be finally obtained by replacing the unknown μ_k in the conditional bias B_c by the estimate $\hat{\mu}_k$. The next subsections give the nonresponse error and the estimator of the conditional bias for each population parameter.

9.2.1 For the generalized total

Using the expression (8.36), the estimate of the conditional bias is

$$\hat{B}_c = W_{x0} + \sum_{k \in S_{I,r}} W_{xk} \hat{\mu}_k - \sum_{k \in S_{I,m}} w_{I,k} \tilde{x}_{I,yk} \hat{\mu}_k. \quad (9.4)$$

9.2.2 For the generalized ratio

Using the expression (8.37), the estimate of the conditional bias is

$$\hat{B}_c = \frac{W_{x0} + \sum_{k \in S_{I,r}} W_{xk} \hat{\mu}_k - \sum_{k \in S_{I,m}} w_{I,k} \tilde{x}_{I,yk} \hat{\mu}_k}{\sum_{k \in S_I} w_{I,k} z_{I,k}}. \quad (9.5)$$

9.2.3 For the domain total

Using the expression (8.38), the estimate of the conditional bias is

$$\hat{B}_c = W_{d0} + \sum_{k \in S_{I,r}} W_{dk} \hat{\mu}_k - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} \hat{\mu}_k. \quad (9.6)$$

9.2.4 For the domain mean

Using the expression (8.39), the estimate of the conditional bias is

$$\hat{B}_c = \frac{W_{d0} + \sum_{k \in S_{I,r}} W_{dk} \hat{\mu}_k - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} \hat{\mu}_k}{\hat{N}_d}. \quad (9.7)$$

9.2.5 For the domain ratio

Using the expression (8.40), the estimate of the conditional bias is

$$\hat{B}_c = \frac{W_{d0} + \sum_{k \in S_{I,r}} W_{dk} \hat{\mu}_k - \sum_{k \in S_{I,m}} w_{I,k} \tilde{d}_{I,yk} \hat{\mu}_k}{\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,zk} z_{I,k}}. \quad (9.8)$$

9.3 Sampling variance

Consider θ , a population parameter of interest, based on the variable of interest y . Let $\hat{\theta}$ be the estimator one would have in the case of complete response. Let $\Psi_s(y)$ an estimator of the variance due to sampling, $\text{var}_{\mathbb{P}}(\hat{\theta})$, that one could use if there were no nonresponse and imputation. The goal is to get an estimate for $\text{var}_{\mathbb{P}}(\hat{\theta})$ when nonresponse and imputation occurred. In the presence of missing values and imputation, $\Psi_s(y) \equiv \hat{V}_{\mathbb{P}}(\hat{\theta})$ cannot be used since the y -values are not observed for the nonrespondents. An asymptotic $\mathbb{MPPQ}^*$ -unbiased estimator $\hat{V}_{\text{SAM}}$ of the sampling variance $V_{\text{SAM}} = E_{\mathbb{M}} \text{var}_{\mathbb{P}}(\hat{\theta})$ can be obtained by finding an asymptotically $\mathbb{M}$ -unbiased estimator for $E_{\mathbb{M}}(\Psi_s(y) | \mathbf{Y}_r)$, where $\mathbf{Y}_r$ is the vector containing the y -values of the respondents.

We will concentrate hereafter on the case of a generalized total (8.6), $T = \sum_{i \in U_S} \tilde{y}_{S,xi}$. The extension to a domain total will be obvious.

The expectation $E_{\mathbb{m}}(\Psi_s(y)|\mathbf{Y}_r)$ depends on the imputation model's parameters, $\tilde{\mu}_{s,yxi} = E_{\mathbb{m}}(\tilde{y}_{s,xi}|\mathbf{Y}_r)$ and $\tilde{\sigma}_{s,yxi}^2 = \text{var}_{\mathbb{m}}(\tilde{y}_{s,xi}|\mathbf{Y}_r)$, which will need to be estimated. We will come back to these quantities later.

For the two-phase sampling, one considers a g-weighted version of the variance estimator (7.3) of Estevao and Särndal (2009). To simplify, we consider all regression coefficients as fixed (with respect to the imputation model). This estimator takes on the form:

$$\begin{aligned}\Psi_s(y, \mathbf{z}) = & \sum_{i \in s_1} \sum_{j \in s_1} D_{1ij} (g_{1i} e_{1i}) (g_{1j} e_{1j}) + \sum_{i \in s} \sum_{j \in s} D_{ij} g_i (y_i - c_i) g_j (y_j - c_j) \\ & + 2 \sum_{i \in s_1} \sum_{j \in s} D_{1ij} a_{2j} g_{1i} e_{1i} g_j (y_j - c_j)\end{aligned}$$

where e_{1i} and c_i are considered fixed⁸ (with respect to the imputation model), as well as D_{1ij} , D_{ij} , a_{2j} , g_{1i} and g_i . The quantity $\mathbf{z}$ combines the information about these quantities considered fixed. To alleviate the notation, the variable $\tilde{y}_{s,xi}$ has been rewritten y_i above and s_1 is the first-phase sample of sampling units, while s is the second-phase sample of sampling units.

We have the three following results:

$$\text{Result 1} \quad E_{\mathbb{m}} \left(\sum_{i \in s_1} \sum_{j \in s_1} D_{1ij} (g_{1i} e_{1i}) (g_{1j} e_{1j}) \mid \mathbf{Y}_r \right) = \sum_{i \in s_1} \sum_{j \in s_1} D_{1ij} (g_{1i} e_{1i}) (g_{1j} e_{1j})$$

$$\begin{aligned}\text{Result 2} \quad E_{\mathbb{m}} \left(\sum_{i \in s} \sum_{j \in s} D_{ij} g_i (y_i - c_i) g_j (y_j - c_j) \mid \mathbf{Y}_r \right) \\ = \sum_{i \in s} \sum_{j \in s} D_{ij} g_i (\tilde{\mu}_i - c_i) g_j (\tilde{\mu}_j - c_j) + \sum_{i \in s} D_{ii} g_i^2 \tilde{\sigma}_i^2\end{aligned}$$

$$\text{Result 3} \quad E_{\mathbb{m}} \left(\sum_{i \in s_1} \sum_{j \in s} D_{1ij} a_{2j} g_{1i} e_{1i} g_j (y_j - c_j) \mid \mathbf{Y}_r \right) = \sum_{i \in s_1} \sum_{j \in s} D_{1ij} a_{2j} g_{1i} e_{1i} g_j (\tilde{\mu}_j - c_j)$$

where $\tilde{\mu}_i$ and $\tilde{\sigma}_i^2$ have been written in place of the more cumbersome notation, $\tilde{\mu}_{s,yxi}$ and $\tilde{\sigma}_{s,yxi}^2$.

The proof of results 1 and 3 is trivial. Result 2 is proved by noting that

- i) $E_{\mathbb{m}} \left((y_i - c_i)(y_j - c_j) \mid \mathbf{Y}_r \right) = (\tilde{\mu}_i - c_i)(\tilde{\mu}_j - c_j), \quad i \neq j;$
- ii) $E_{\mathbb{m}} \left((y_i - c_i)^2 \mid \mathbf{Y}_r \right) = \text{var}_{\mathbb{m}} \left((y_i - c_i) \mid \mathbf{Y}_r \right) + \left[E_{\mathbb{m}} \left((y_i - c_i) \mid \mathbf{Y}_r \right) \right]^2 = \tilde{\sigma}_i^2 + (\tilde{\mu}_i - c_i)^2.$

⁸ The quantities e_{1i} and c_j are of the form of a beta coefficient and in that sense convergence theorems under mild regularity conditions allow us to consider them as fixed under the imputation model.

Combining these three results and noting that $D_{ii} = (1 - \pi_i)/\pi_i^2$, where π_i is the product of selection probabilities in each phase, we get

$$E_{\mathbb{m}}(\Psi_s(y, \mathbf{z}) | \mathbf{Y}_r) = \Psi_s(\tilde{\mu}, \mathbf{z}) + \sum_{i \in S} (1 - \pi_i) w_i^2 \tilde{\sigma}_i^2 \quad (9.9)$$

where w_i is the estimation weight of unit i .

The expression above is not an estimator, as the following quantities are not known, either for all units or for nonrespondents: $\tilde{\mu}_i = \tilde{\mu}_{S,yxi}$, $\tilde{\sigma}_i^2 = \tilde{\sigma}_{S,yxi}^2$, $\mathbf{z}_i$ (indeed the variables in $\mathbf{z}$ involve derivations based on the y -values of all units, which we don't have for nonrespondents).

In the case of $\tilde{\mu}_i$ and $\tilde{\sigma}_i^2$, they can be estimated in the following way. Using (8.11), we have $\tilde{y}_{S,xi} = \sum_{k \in U_{I,i}} \tilde{x}_{I,yk} y_{I,k}$, where $U_{I,i}$ is the set of imputation units within the sampling unit i and $\tilde{x}_{I,yk}$ is fixed with respect to the imputation model. It is noteworthy that some imputation units k can be imputed and others can be responded within the same sampling unit i .

We make the important following assumption: the imputation units are independent under the imputation model. This allows us to write

$$\begin{aligned} \tilde{\mu}_{S,yxi} &= E_{\mathbb{m}}(\tilde{y}_{S,xi} | \mathbf{Y}_r) = \sum_{k \in U_{I,i}} \tilde{x}_{I,yk} [r_{yk} y_{ik} + (1 - r_{yk}) \mu_{yk}], \\ \tilde{\sigma}_{S,yxi}^2 &= \text{var}_{\mathbb{m}}(\tilde{y}_{S,xi} | \mathbf{Y}_r) = \sum_{k \in U_{I,i}} (1 - r_{yk}) \tilde{x}_{I,yk}^2 \sigma_{yk}^2, \end{aligned}$$

where $r_{yk} = 1$ if the imputation unit k is a respondent for y , and $r_{yk} = 0$ if it is a nonrespondent for y . Quantities μ_{yk} and σ_{yk}^2 are defined in (7.1) and (7.2), respectively.

If all imputation units within the sampling unit i are respondents ($r_{yk} = 1$ for all $k \in U_{I,i}$), then $\tilde{\mu}_{S,yxi} = y_{S,yxi}$ and $\tilde{\sigma}_{S,yxi}^2 = 0$.

The estimates for $\tilde{\mu}_{S,yxi}$ and $\tilde{\sigma}_{S,yxi}^2$ are thus

$$\hat{\mu}_{S,yxi} = \sum_{k \in U_{I,i}} \tilde{x}_{I,yk} [r_{yk} y_{I,k} + (1 - r_{yk}) \hat{\mu}_{yk}] \quad (9.10)$$

and

$$\hat{\sigma}_{S,yxi}^2 = \sum_{k \in U_{I,i}} (1 - r_{yk}) \tilde{x}_{I,yk}^2 \hat{\sigma}_{yk}^2, \quad (9.11)$$

where $\hat{\mu}_{yk}$ and $\hat{\sigma}_{yk}^2$ are estimates for μ_{yk} and σ_{yk}^2 respectively (see chapter 7).

The quantities in $\mathbf{z}$ that depend on the y -values are linear expressions in y (for instance, $\hat{\beta}$ in a regression is linear in y). Therefore, they can be estimated by replacing $y_{S,yxi}$ by $\hat{\mu}_{S,yxi}$ in their linear expressions, using the Continuous Mapping Theorem to justify this substitution. Therefore the quantities $\mathbf{z}_i$, which at the beginning were derived from the y -values, are estimated by $\hat{\mathbf{z}}_i$, which can be derived from the $\hat{\mu}_{S,yx}$ -values. This entails that the estimator of the sampling variance in presence of imputation is

$$\hat{V}_{\text{SAM}} = \Psi(\hat{\mu}_{S,yx}, \hat{\mathbf{z}}) + \sum_{i \in S} (1 - \pi_i) w_i^2 \hat{\sigma}_{S,yxi}^2$$

which we rewrite as

$$\hat{V}_{\text{SAM}} = \Psi(\hat{\mu}_{S,yx}) + \sum_{i \in S} (1 - \pi_i) w_i^2 \hat{\sigma}_{S,yxi}^2$$

where the omission of the second argument, $\hat{\mathbf{z}}$, means that this second argument is derived from the first argument.

It is noteworthy that the estimator, $\hat{\theta}_\bullet$, is based on the $y_{\bullet i}$ -values, but the estimator of the sampling variance is based on the $\hat{\mu}_{S,yxi}$ -values.

The derivation of $\hat{V}_{\text{SAM}}$ was done for the two-phase sampling, but it covers as a special case the one-phase sampling also. Also, going from a generalized total to a domain total (8.8), the following quantities will be required in place of $\hat{\mu}_{S,yxi}$ and $\hat{\sigma}_{S,yxi}^2$:

$$\hat{\mu}_{S,ydi} = \sum_{k \in U_{I,i}} \tilde{d}_{I,yk} [r_{yk} y_{I,k} + (1 - r_{yk}) \hat{\mu}_{yk}] \quad (9.12)$$

and

$$\hat{\sigma}_{S,ydi}^2 = \sum_{k \in U_{I,i}} (1 - r_{yk}) \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2. \quad (9.13)$$

Finally, the cases of a generalized ratio, a domain mean or a domain ratio are obtained by linearization techniques.

9.3.1 For the generalized total

$$\hat{V}_{\text{SAM}} = \Psi(\hat{\mu}_{S,yx}, \text{GTOTAL}) + \sum_{i \in S} (1 - \pi_i) w_{S,i}^2 \hat{\sigma}_{S,yxi}^2$$

where Ψ is the functional providing a sampling variance in the case of a generalized total.

9.3.2 For the generalized ratio

$$\hat{V}_{SAM} = \Psi(\hat{\mu}_{S,yx}, \text{GRATIO}) + \frac{\sum_{i \in S_S} (1 - \pi_i) w_{S,i}^2 \hat{\sigma}_{S,yxi}^2}{\left(\sum_{j \in S_E} w_{E,j} z_{E,j} \right)^2}$$

where Ψ is the functional providing a sampling variance estimate in the case of a generalized ratio.

9.3.3 For the domain total

$$\hat{V}_{SAM} = \Psi(\hat{\mu}_{S,yd}, \text{DTOTAL}) + \sum_{i \in S_S} (1 - \pi_i) w_{S,i}^2 \hat{\sigma}_{S,ydi}^2$$

where Ψ is the functional providing a sampling variance estimate in the case of a domain total.

9.3.4 For the domain mean

$$\hat{V}_{SAM} = \Psi(\hat{\mu}_{S,yd}, \text{DMEAN}) + \frac{\sum_{i \in S_S} (1 - \pi_i) w_{S,i}^2 \hat{\sigma}_{S,ydi}^2}{[\hat{N}_d]^2}$$

where Ψ is the functional providing a sampling variance estimate in the case of a domain mean.

9.3.5 For the domain ratio

$$\hat{V}_{SAM} = \Psi(\hat{\mu}_{S,yd}, \text{RATIO}, z) + \frac{\sum_{i \in S_S} (1 - \pi_i) w_{S,i}^2 \hat{\sigma}_{S,ydi}^2}{\left[\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j} \right]^2}$$

where Ψ is the functional providing a sampling variance estimate in the case of a domain ratio.

9.4 Nonresponse variance

An estimator $\hat{V}_{NR}$ of the nonresponse variance $V_{NR} = E_{\mathbb{P}_{\mathbb{Q}}} E_{\mathbb{M}^*} [(\hat{\theta}_{\bullet} - \hat{\theta})^2 | s, s_r]$ can simply be obtained by estimating $E_{\mathbb{M}^*} [(\hat{\theta}_{\bullet} - \hat{\theta})^2 | s, s_r]$. The latter can be expressed as:

$$\begin{aligned} E_{\mathbb{M}^*} [(\hat{\theta}_{\bullet} - \hat{\theta})^2 | s, s_r] &= \text{var}_{\mathbb{M}^*} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r] + B_c^2 \\ &= \text{var}_{\mathbb{M}} E_{\star} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r, \mathbf{Y}] + E_{\mathbb{M}} \text{var}_{\star} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r, \mathbf{Y}] + B_c^2, \end{aligned}$$

where $\mathbf{Y}$ is the vector of population y -values and $B_c = E_{\mathbb{M}^*} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r]$ is the conditional bias (see section 9.2). The squared conditional bias B_c^2 is estimated by $\hat{B}_c^2$.

The quantity $E_{\mathbb{M}} \text{var}_{\star} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r, \mathbf{Y}]$ can be viewed as the additional component of the nonresponse variance that is due to using a random imputation mechanism, if any. This additional component is sometimes called the imputation variance. A natural $\mathbb{M}^*$ -unbiased estimator of the imputation variance is $\text{var}_{\star} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r, \mathbf{Y}]$. It is obtained from the expressions for the nonresponse error given in section 8.4.3 and the description of stochastic imputation in section 6.4.

The third and last term to be estimated, $\text{var}_{\mathbb{M}} E_{\star} [(\hat{\theta}_{\bullet} - \hat{\theta}) | s, s_r, \mathbf{Y}]$, is also obtained from the expressions for the nonresponse error and the description of stochastic imputation. It is estimated by replacing the unknown $\sigma_{I,k}^2$ by $\hat{\sigma}_{I,k}^2$. The resulting nonresponse variance estimator $\hat{V}_{NR}$ is provided in the following subsections.

9.4.1 For the generalized total

$$\hat{V}_{NR} = \sum_{k \in S_{I,r}} (W_{I,xk})^2 \hat{\sigma}_{I,k}^2 + \sum_{k \in S_{I,m}} (1 + I_{y,\text{RESID},k}) w_{I,k}^2 \tilde{x}_{I,yk}^2 \hat{\sigma}_{I,k}^2 + \hat{B}_c^2.$$

9.4.2 For the generalized ratio

$$\hat{V}_{NR} = \frac{\sum_{k \in S_{I,r}} (W_{I,xk})^2 \hat{\sigma}_{I,k}^2 + \sum_{k \in S_{I,m}} (1 + I_{y,\text{RESID},k}) w_{I,k}^2 \tilde{x}_{I,yk}^2 \hat{\sigma}_{I,k}^2}{(\sum_{k \in S_I} w_{I,k} z_{I,k})^2} + \hat{B}_c^2.$$

9.4.3 For the domain total

$$\hat{V}_{\text{NR}} = \sum_{k \in S_{I,y,r}} (W_{I,dk})^2 \hat{\sigma}_k^2 + \sum_{k \in S_{I,y,m}} (1 + I_{y,\text{RESID},k}) w_{I,k}^2 \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2 + \hat{B}_c^2.$$

9.4.4 For the domain mean

$$\hat{V}_{\text{NR}} = \frac{\sum_{k \in S_{I,y,r}} (W_{I,dk})^2 \hat{\sigma}_k^2 + \sum_{k \in S_{I,y,m}} (1 + I_{y,\text{RESID},k}) w_{I,k}^2 \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2}{(\hat{N}_d)^2} + \hat{B}_c^2.$$

9.4.5 For the domain ratio

$$\hat{V}_{\text{NR}} = \frac{\sum_{k \in S_{I,y,r}} (W_{I,dk})^2 \hat{\sigma}_k^2 + \sum_{k \in S_{I,y,m}} (1 + I_{y,\text{RESID},k}) w_{I,k}^2 \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2}{(\sum_{j \in S_E} w_{E,j} I_{dj} z_{E,j})^2} + \hat{B}_c^2.$$

Note that the denominator above is equal to $(\sum_{k \in S_I} w_{I,k} \tilde{d}_{I,zk} z_{I,k})^2$.

9.5 Mixed component estimation

An estimator $\hat{V}_{\text{MIX}}$ of $V_{\text{MIX}} = 2E_{\mathbb{P}_{\text{q}}} E_{\text{mm}^*}[(\hat{\theta} - \theta)(\hat{\theta}_\bullet - \hat{\theta})|s, s_r]$ is obtained by estimating $2E_{\text{mm}^*}[(\hat{\theta} - \theta)(\hat{\theta}_\bullet - \hat{\theta})|s, s_r]$. Assuming that the conditional bias $B_c = E_{\text{mm}^*}[(\hat{\theta}_\bullet - \hat{\theta})|s, s_r]$ is negligible, we obtain the following approximation:

$$\begin{aligned} 2E_{\text{mm}^*}[(\hat{\theta} - \theta)(\hat{\theta}_\bullet - \hat{\theta})|s, s_r] &\approx 2 \text{cov}_{\text{mm}^*}[(\hat{\theta} - \theta), (\hat{\theta}_\bullet - \hat{\theta})|s, s_r] \\ &= 2 \text{cov}_{\text{mm}}\{(\hat{\theta} - \theta), E_*[(\hat{\theta}_\bullet - \hat{\theta})|s, s_r, \mathbf{Y}]|s, s_r\} \end{aligned}$$

Greater justification of this approximation is given in Beaumont and Bissonnette (2011). Unlike the estimation of the nonresponse variance in section 9.4, note that there is no extra component due to using a random imputation mechanism. To obtain an estimator of the mixed component, we use the decompositions of the sampling error, $\hat{\theta} - \theta$, from section 8.4.2. From the form of this decomposition, the expression of the nonresponse error (section 8.4.3) and the description of stochastic imputation (section 6.4), it is straightforward to obtain an expression for

$$2 \text{cov}_{\text{mm}}\{(\hat{\theta} - \theta), E_*[(\hat{\theta}_\bullet - \hat{\theta})|s, s_r, \mathbf{Y}]|s, s_r\}.$$

This expression involves an unknown ratio: ρ_z , ρ_d or ρ_{zd} , if the parameter of interest is a generalized ratio, a domain mean or a domain ratio (section 8.4.2). Since this unknown ratio converges in probability to 1, we will approximate it by 1. The above expression also involves the unknown σ_k^2 , which will be estimated by $\hat{\sigma}_k^2$. After these substitutions, we get the estimator $\hat{V}_{\text{MIX}}$ of the mixed component, as given in the next subsections.

9.5.1 For the generalized total

$$\hat{V}_{\text{MIX}} = 2 \sum_{k \in S_{I,y,r}} (w_{I,k} - 1) W_{ydk} \tilde{x}_{I,yk} \hat{\sigma}_{yk}^2 - 2 \sum_{k \in S_{I,y,m}} w_{I,k} (w_{I,k} - 1) \tilde{x}_{I,yk}^2 \hat{\sigma}_{yk}^2, \quad (9.14)$$

9.5.2 For the generalized ratio

$$\hat{V}_{T,\text{MIX}} = \frac{2 \sum_{k \in S_{I,y,r}} (w_{I,k} - 1) W_{ydk} \tilde{x}_{I,yk} \hat{\sigma}_{yk}^2 - 2 \sum_{k \in S_{I,y,m}} w_{I,k} (w_{I,k} - 1) \tilde{x}_{I,yk}^2 \hat{\sigma}_{yk}^2}{\left(\sum_{j \in S_E} w_{E,j} z_{E,j} \right)^2} \quad (9.15)$$

9.5.3 For the domain total

$$\hat{V}_{T,\text{MIX}} = 2 \sum_{k \in S_{I,y,r}} (w_{I,k} - 1) W_{ydk} \tilde{d}_{I,yk} \hat{\sigma}_{yk}^2 - 2 \sum_{k \in S_{I,y,m}} w_{I,k} (w_{I,k} - 1) \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2 \quad (9.16)$$

9.5.4 For the domain mean

$$\hat{V}_{\text{MIX}} = \frac{2 \sum_{k \in S_{I,y,r}} (w_{I,k} - 1) W_{ydk} \tilde{d}_{I,yk} \hat{\sigma}_{yk}^2 - 2 \sum_{k \in S_{I,y,m}} w_{I,k} (w_{I,k} - 1) \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2}{(\tilde{N}_d)^2} \quad (9.17)$$

9.5.5 For the domain ratio

$$\hat{V}_{\text{MIX}} = \frac{2 \sum_{k \in S_{I,y,r}} (w_{I,k} - 1) W_{ydk} \tilde{d}_{I,yk} \hat{\sigma}_{yk}^2 - 2 \sum_{k \in S_{I,y,m}} w_{I,k} (w_{I,k} - 1) \tilde{d}_{I,yk}^2 \hat{\sigma}_{yk}^2}{\left(\sum_{k \in S_E} w_{E,j} I_{dj} Z_{E,j} \right)^2} \quad (9.18)$$

9.6 Some remarks

9.6.1 Relevancy of μ_k and σ_k^2

All the estimates described in the previous sections, for the conditional bias, the sampling variance, the nonresponse variance and the sampling-nonresponse covariance, make use of $\hat{\mu}_k$'s or $\hat{\sigma}_k^2$'s derived in chapter 7, Imputation model.

For a respondent k , this value of $\hat{\mu}_k$ or $\hat{\sigma}_k^2$ is always multiplied by the compensation weight, $W_{I,xk}$ or $W_{I,dk}$ (derived in (8.29)).

It is interesting to see that the value of $\hat{\mu}_k$ or $\hat{\sigma}_k^2$ for a respondent k only matters if this respondent contributed to the imputation of at least one nonrespondent. Indeed, only in that case the value of the compensation weight, $W_{I,xk}$ or $W_{I,dk}$, can have a nonzero value. So, for a variable of interest y , we can divide the units of the sample into three groups: the nonrespondents in y , the respondents in y that contributed to imputation of some nonrespondents and the respondents in y that did not contribute to imputation of any nonrespondent. Only the $\hat{\mu}_k$'s and $\hat{\sigma}_k^2$'s of the first two groups matter.

9.7 Total variance

The estimate of the total variance for the estimator $\hat{\theta}_\bullet$ is

$$\hat{V}_{\text{TOT}} = \hat{V}_{\text{SAM}} + \hat{V}_{\text{NR}} + \hat{V}_{\text{MIX}} .$$

10 Summary for the simple case

This section is a summary of all the sections above, except that it corresponds to the following simple case:

- One sampling unit equals one imputation unit equals one estimation unit. We will simply talk of *units*, without stating imputation or sampling or estimation in from of it.
- No imputation step uses an adjustment variable x_a .
- No imputation step uses a variance structure variable v .
- The only population parameters of interest are domain totals, domain means and domain ratios.
- No random residual is added to the imputed values.

Here we will use U to denote the population and s to denote the sample. The index k will represent a unit.

Consider a variable of interest y .

In the remaining of this section, we will work only with that variable y . Therefore the ensuing notation will omit a subscript for y but it is implicit.

Let s_r ($=s_{yr}$) be the part of the sample where y has been responded and s_m ($=s_{ym}$) the part where it has been nonresponded and then imputed. Say that J ($=J_y$) imputation steps were required to impute all nonrespondents in terms of the variable y . Each of these J steps used one of the imputation methods described in the next section.

10.1 Imputation methods

10.1.1 Auxiliary value imputation

y_l is said to be imputed by auxiliary value (AV) imputation if the imputed value y_l^* is calculated as

$$y_l^* = x_l$$

for some variable x . This method of imputation only uses data from the same record as the nonrespondent.

10.1.2 Donor imputation

y_l is said to be imputed by donor if the imputed value y_l^* is

$$y_l^* = y_k$$

for some $k \in s_r$. In other words, the value imputed to y_l is the direct value of a respondent. The choice of the respondent may depend on matching fields (nearest donor) or may be entirely random (random donor).

10.1.3 Linear regression imputation

y_l is said to be imputed by linear regression (LR) if the imputed value y_l^* is calculated as

$$y_l^* = \left[\left(\sum_{k \in s_r} \mathbf{x}_k \mathbf{x}_k' \right)^{-1} \sum_{k \in s_r} \mathbf{x}_k y_k \right] \mathbf{x}_l$$

for some vector of auxiliary variables $\mathbf{x}$.

10.1.4 Linear imputation

Linear imputation is not a fourth imputation method. It is instead an umbrella method that contains the three previous imputation methods as specific cases.

We say that an imputation method is linear if the imputed value y_l^* can be expressed as a linear combination of the values of y of the respondents:

$$y_l^* = \varphi_{0l} + \sum_{k \in s_r} \varphi_{kl} y_k$$

where the quantities φ_{0l} and φ_{kl} can depend on the full sample for auxiliary variables, but do not depend at all on the y values.

10.1.5 All J imputation steps

Variable y has needed J imputation steps to finish all imputation. It means that the set of nonrespondents s_m is partitionned into J subsets: $s_m = \bigcup_{j=1}^J s_m^{(j)}$, where $s_m^{(j)}$ ($=s_{ym}^{(j)}$) is the set of nonrespondents that have been imputed at imputation step number j .

Imputation at step number j can be written as

$$y_l^* = \varphi_{0l}^{(j)} + \sum_{k \in s_r} \varphi_{kl}^{(j)} y_k, \quad l \in s_m^{(j)}.$$

Consider now some $l \in s_m$. Then there is one and only one index j such that $l \in s_m^{(j)}$. For this unit l , the quantities $\varphi_{0l}^{(j)}$ and $\varphi_{kl}^{(j)}$ need to be defined, for all $k \in s_r$, but the quantities $\varphi_{0l}^{(j')}$ and $\varphi_{kl}^{(j')}$ need not, when $j' \neq j$. Therefore we can define φ_{0l} and φ_{kl} as:

$$\varphi_{0l} = \begin{cases} \varphi_{0l}^{(1)}, & l \in s_m^{(1)} \\ \varphi_{0l}^{(2)}, & l \in s_m^{(2)} \\ \vdots & \vdots \\ \varphi_{0l}^{(j)}, & l \in s_m^{(j)} \end{cases},$$

$$\varphi_{kl} = \begin{cases} \varphi_{kl}^{(1)}, & l \in s_m^{(1)} \\ \varphi_{kl}^{(2)}, & l \in s_m^{(2)} \\ \vdots & \vdots \\ \varphi_{kl}^{(j)}, & l \in s_m^{(j)} \end{cases}, k \in s_r.$$

In short, we can state that the imputed value of y_l , for $l \in s_m$, is

$$y_l^* = \varphi_{0l} + \sum_{k \in s_r} \varphi_{kl} y_k, \quad l \in s_m.$$

10.2 Conditional moments under the imputation model

For each imputation step there is an underlying imputation model, say $\mathbb{m}$, based on the imputation method.

Let

$$\mu_k = E_{\mathbb{m}}(y_k | \mathbf{x}_k),$$

$$\sigma_k^2 = \text{var}_{\mathbb{m}}(y_k | \mathbf{x}_k),$$

where the subscript $\mathbb{m}$ indicates that the expectation is taken under the imputation model and $\mathbf{x}$ is the vector of auxiliary variables used for the imputation. Let $\hat{\mu}_k$ and $\hat{\sigma}_k^2$ be the estimates for μ_k and σ_k^2 .

10.3 Population parameters

Let d be the index of a specific domain. Let defined $I_{dk} = 1$ if unit k belongs to the domain number d , and 0 otherwise. The knowledge about d must be available for all units in the sample and not be subject to nonresponse.

10.3.1 Domain total

$$T_{yd} = \sum_{k \in U} I_{dk} y_k . \quad (10.1)$$

10.3.2 Domain mean

$$M_{yd} = \sum_{k \in U} I_{dk} y_k / \sum_{k \in U} I_{dk} . \quad (10.2)$$

10.3.3 Domain ratio

$$R_{y zd} = \frac{\sum_{k \in U} I_{dk} y_k}{\sum_{k \in U} I_{dk} z_k} . \quad (10.3)$$

The variable z at the denominator must be fully responded in the sample (no nonresponse or imputation).

10.4 Estimators under full response

Let w_k be the estimation weight of unit $k \in s$.

If there was no nonresponse, the estimators used would be:

10.4.1 Domain total

$$\hat{T}_{yd} = \sum_{k \in s} w_k I_{dk} y_k . \quad (10.4)$$

10.4.2 Domain mean

$$\hat{M}_{yd} = \sum_{k \in S} w_k I_{dk} y_k / \sum_{k \in S} w_k I_{dk} . \quad (10.5)$$

10.4.3 Domain ratio

$$\hat{R}_{ydz} = \frac{\sum_{k \in S} w_k I_{dk} y_k}{\sum_{k \in S} w_k I_{dk} z_k} . \quad (10.6)$$

10.5 Estimators under nonresponse and imputation

Let

$$y_{\bullet k} = \begin{cases} y_k, & k \in s_r \\ y_k^*, & k \in s_m \end{cases} .$$

This is the value available for variable y and unit k .

When there is nonresponse and imputation on y , the estimators, sometimes called the imputed estimators, are:

10.5.1 Domain total

$$\hat{T}_{\bullet yd} = \sum_{k \in S} w_k I_{dk} y_{\bullet k} . \quad (10.7)$$

10.5.2 Domain mean

$$\hat{M}_{\bullet yd} = \sum_{k \in S} w_k I_{dk} y_{\bullet k} / \sum_{k \in S} w_k I_{dk} . \quad (10.8)$$

10.5.3 Domain ratio

$$\hat{R}_{\bullet yzd} = \frac{\sum_{k \in S} w_k I_{dk} y_{\bullet k}}{\sum_{k \in S} w_k I_{dk} z_k} . \quad (10.9)$$

10.6 Canonical form of the estimators

Let

$$W_{d0} = \sum_{l \in S_m} w_l I_{dl} \varphi_{0l},$$
$$W_{dk} = \sum_{l \in S_m} w_l I_{dl} \varphi_{kl} , \quad k \in S_r .$$

Then the estimators under imputation can be expressed as:

10.6.1 Domain total

$$\hat{T}_{\bullet yd} = W_{d0} + \sum_{k \in S_r} (w_k I_{dk} + W_{dk}) y_k . \quad (10.10)$$

10.6.2 Domain mean

$$\hat{M}_{\bullet yd} = \frac{W_{d0} + \sum_{k \in S_r} (w_k I_{dk} + W_{dk}) y_k}{\sum_{k \in S} w_k I_{dk}} . \quad (10.11)$$

10.6.3 Domain ratio

$$\hat{R}_{\bullet yzd} = \frac{W_{d0} + \sum_{k \in S_r} (w_k I_{dk} + W_{dk}) y_k}{\sum_{k \in S} w_k I_{dk} z_k} . \quad (10.12)$$

10.7 Bias

10.7.1 Domain total

$$\hat{B}_c = W_{d0} + \sum_{k \in S_r} W_{dk} \hat{\mu}_k - \sum_{k \in S_m} w_k I_{dk} \hat{\mu}_k. \quad (10.13)$$

10.7.2 Domain mean

$$\hat{B}_c = \frac{W_{d0} + \sum_{k \in S_r} W_{dk} \hat{\mu}_k - \sum_{k \in S_m} w_k I_{dk} \hat{\mu}_k}{\sum_{k \in S} w_k I_{dk}}. \quad (10.14)$$

10.7.3 Domain ratio

$$\hat{B}_c = \frac{W_{d0} + \sum_{k \in S_r} W_{dk} \hat{\mu}_k - \sum_{k \in S_m} w_k I_{dk} \hat{\mu}_k}{\sum_{k \in S} w_k I_{dk} z_k}. \quad (10.15)$$

10.8 Sampling variance

Let

$$\tilde{y}_k = \begin{cases} y_k, & k \in s_r \\ \hat{\mu}_k, & k \in s_m \end{cases}.$$

For the sampling design of the sample, the type of estimator (Horvitz-Thompson estimator or calibrated estimator) and the population parameter, there is a function Ψ providing an estimate for the sampling variance of the estimator under full response.

For example, if (a) the design is a stratified simple random sampling without replacement, (b) the estimator is the Horvitz-Thompson estimator and (c) the population parameter is the total of a variable u , then the function Ψ takes the following form:

$$\Psi(u) = \sum_{h=1}^H N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_{uh}^2}{n_h},$$

where $s_{uh}^2 = \frac{1}{n_h-1} \sum_{k \in s_h} (u_k - \bar{u}_h)^2$ and $\bar{u}_h = \frac{1}{n_h} \sum_{k \in s_h} u_k$.

In other cases, the function Ψ may use linearization and/or residuals. But it is assumed that such a formula for Ψ is available for fully responded variables.

Under nonresponse and imputation, we cannot use directly the function Ψ on the $y_{\bullet k}$ values. However the sampling variance for the imputed estimator will nevertheless make use in some way of the function Ψ , except that it will be on a slightly modified variable ($\tilde{y}$ instead of $y_{\bullet}$) and an additional term will need to be added to the sampling variance.

In the following, π_k is the selection probability of unit $k \in s$. In the case of a two-phase design, we define π_k as the product of the first-phase selection probability and the conditional selection probability at the second phase.

10.8.1 Domain total

$$\hat{V}_{SAM} = \Psi(\tilde{y}) + \sum_{k \in s_m} (1 - \pi_k) w_k^2 I_{dk} \hat{\sigma}_k^2$$

where $\Psi(u)$ is the function providing a sampling variance estimate for the domain total of a fully responded variable u .

Please remark that if the variable y was fully responded, then $\tilde{y}_k$ is systematically equal to y_k for all $k \in s$ and $s_m = \emptyset$, and therefore

$$\hat{V}_{SAM} = \Psi(y) + 0 = \Psi(y)$$

which is the estimator in the situation of full response.

10.8.2 Domain mean

$$\hat{V}_{SAM} = \Psi(\tilde{y}) + \frac{\sum_{k \in s_m} (1 - \pi_k) w_k^2 I_{dk} \hat{\sigma}_k^2}{\left(\sum_{k \in s} w_k I_{dk} \right)^2}$$

where $\Psi(u)$ is the function providing a sampling variance estimate for the domain mean of a fully responded variable u .

The remark done for the domain total still applies here: if y was fully responded, the sampling variance estimator above falls back to the traditional estimator $\Psi(y)$.

10.8.3 Domain ratio

$$\hat{V}_{SAM} = \Psi(\tilde{y}, z) + \frac{\sum_{k \in S_m} (1 - \pi_k) w_k^2 I_{dk} \hat{\sigma}_k^2}{\left(\sum_{k \in S} w_k I_{dk} z_k \right)^2}$$

where $\Psi(u, v)$ is the function providing a sampling variance estimate for the domain ratio of two fully responded variables u and v .

The remark done for the domain total still applies here: if y was fully responded, the sampling variance estimator above falls back to the traditional estimator $\Psi(y)$.

10.9 Nonresponse variance

10.9.1 Domain total

$$\hat{V}_{NR} = \sum_{k \in S_r} W_{dk}^2 \hat{\sigma}_k^2 + \sum_{k \in S_m} w_k^2 I_{dk} \hat{\sigma}_{yk}^2 + \hat{B}_c^2.$$

10.9.2 Domain mean

$$\hat{V}_{NR} = \frac{\sum_{k \in S_r} W_{dk}^2 \hat{\sigma}_k^2 + \sum_{k \in S_m} w_k^2 I_{dk} \hat{\sigma}_{yk}^2 + \hat{B}_c^2}{\left(\sum_{k \in S} w_k I_{dk} \right)^2} + \hat{B}_c^2.$$

10.9.3 Domain ratio

$$\hat{V}_{NR} = \frac{\sum_{k \in S_r} W_{dk}^2 \hat{\sigma}_k^2 + \sum_{k \in S_m} w_k^2 I_{dk} \hat{\sigma}_{yk}^2 + \hat{B}_c^2}{\left(\sum_{k \in S} w_k I_{dk} z_k \right)^2} + \hat{B}_c^2.$$

10.10 Mixed component variance

10.10.1 Domain total

$$\hat{V}_{\text{MIX}} = 2 \sum_{k \in S_r} (w_k - 1) W_{dk} \hat{\sigma}_k^2 - 2 \sum_{k \in S_m} w_k (w_k - 1) I_{dk} \hat{\sigma}_k^2$$

10.10.2 Domain mean

$$\hat{V}_{\text{MIX}} = \frac{2 \sum_{k \in S_r} (w_k - 1) W_{dk} \hat{\sigma}_k^2 - 2 \sum_{k \in S_m} w_k (w_k - 1) I_{dk} \hat{\sigma}_k^2}{\left(\sum_{k \in S} w_k I_{dk} \right)^2}$$

10.10.3 Domain ratio

$$\hat{V}_{\text{MIX}} = \frac{2 \sum_{k \in S_r} (w_k - 1) W_{dk} \hat{\sigma}_k^2 - 2 \sum_{k \in S_m} w_k (w_k - 1) I_{dk} \hat{\sigma}_k^2}{\left(\sum_{k \in S} w_k I_{dk} z_k \right)^2}$$

10.11 Total variance

The total variance estimator is simply the sum of the three components of variance estimates defined above:

$$\hat{V} = \hat{V}_{\text{SAM}} + \hat{V}_{\text{NR}} + \hat{V}_{\text{MIX}}.$$

11References

Binder, D.A. (1983). On the variances of asymptotically normal estimators from complex surveys. *International Statistical Review*, 51, 279-292.

Beaumont, J.-F., and Bissonnette, J. (2011). Variance estimation under composite imputation: the methodology behind SEVANI. *Survey Methodology*, 37, 171-179.

- Beaumont, J.-F., and Bocci, C. (2009). Variance estimation when donor imputation is used to fill in missing values. *Canadian Journal of Statistics*, 37, 400-416.
- Beaumont, J.-F., Haziza, D., and Bocci, C. (2011). On variance estimation under auxiliary value imputation in sample surveys. *Statistica Sinica*, 2, 515-537.
- Brick, J.M., Kalton, G., and Kim, J.K. (2004). Variance estimation with hot deck imputation using a model. *Survey Methodology*, 30, 57-66.
- Deville, J.-C., and Särndal, C.-E. (1992). Calibration estimators in survey sampling. *Journal of the American Statistical Association*, 87, 376-382.
- Deville, J.-C., and Särndal, C.-E. (1994). Variance estimation for the regression imputed Horvitz-Thompson estimator. *Journal of Official Statistics*, 10, 381-394.
- Estevao, V. and Särndal, C.-E. (2009). A new face on two-phase sampling with calibration estimators. *Survey Methodology*, 35, 3-14.
- Särndal, C.-E. (1992). Methods for estimating the precision of survey estimates when imputation has been used. *Survey Methodology*, 18, 241-252.
- SAS Institute Inc. (2004). *SAS 9.1.3, Help and Documentation*. SAS Institute Inc., Cary, NC, U.S.A.
- Statistics Canada (2010). SEVANI version 2.3 – Methodology Guide.